Diagnostic evidence and solutions

Aleph Platinum · Tutor book · Not for learner distribution before completion

Read the separate administration guide and verifier rubric before using these solutions. Every item is original. The level labels are intended difficulty, not calibrated AMC/AIME equivalences. No overall math-ability score is justified.

Day 1: Broad starting sample

Core: A1-1, A3-1, I1-1, F1-1, L1-1, S1-1, C1-1, N3-1. Reserve selection: A1-R, A1-4, A3-R, A3-4, I1-R, I1-4, F1-R, F1-4, L1-R, L1-4, S1-R, S1-4, C1-R, C1-4, N3-R, N3-4.

A1-1 · Equivalent equations and domain · L1

Question: For real x, explain why dividing both sides of an equation by x can lose a solution. Give one example.

Answer: Division requires $x \neq 0$; $x^2 = x$ loses $x = 0$ if divided by x.

1. Division by zero is undefined.
2. Factor $x^2 - x = x(x - 1)$; its roots are 0 and 1.
3. Dividing by x gives only $x = 1$, so treat $x = 0$ separately.

Evidence anchors (0–3, descriptive only):

- 1 point: Division by zero is undefined.
- 1 point: Factor $x^2 - x = x(x - 1)$; its roots are 0 and 1.
- 1 point: give Division requires $x \neq 0$; $x^2 = x$ loses $x = 0$ if divided by x. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

A3-1 · Vieta formulas and symmetric root expressions · L1

Question: If r and s are the roots of $ax^2 + bx + c=0$, $a \neq 0$, what are r+s and rs? Explain by expanding.

Answer: $r+s=-b/a$; $rs=c/a$.

1. Expand $a(x - r)(x - s)=ax^2 - a(r+s)x + ars$.
2. Match coefficients with $ax^2 + bx + c$.

Evidence anchors (0–3, descriptive only):

- 1 point: Expand $a(x - r)(x - s) = ax^2 - a(r+s)x + ars$.
- 1 point: Match coefficients with $ax^2 + bx + c$.
- 1 point: give $r+s=-b/a$; $rs=c/a$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand $(x-r)(x-s)$, or expand a square of a sum.

Follow-up: If the reasoning is uncertain, use A3-R without revealing this solution. If independent reasoning is sound, consider A3-4. Do not infer a gap from an unreached item.

I1-1 · Signs and inequality transformations · L1

Question: What happens to an inequality when both sides are multiplied by a negative number? Give a numerical example.

Answer: The direction reverses; $2<3$ becomes $-2>-3$ after multiplication by -1.

1. Multiplication by a negative reverses order on the real line.
2. The example must reverse the direction, not merely negate the numbers.

Evidence anchors (0–3, descriptive only):

- 1 point: Multiplication by a negative reverses order on the real line.
- 1 point: The example must reverse the direction, not merely negate the numbers.
- 1 point: give The direction reverses; $2<3$ becomes $-2>-3$ after multiplication by -1. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Failing to reverse order after negative division; multiplying by an expression of unknown sign
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Mark points where a factor is zero or undefined. Test one point in each interval.

Follow-up: If the reasoning is uncertain, use I1-R without revealing this solution. If independent reasoning is sound, consider I1-4. Do not infer a gap from an unreached item.

F1-1 · Functions, domain and composition · L1

Question: What does $(f \circ g)(x)$ mean, and which inputs are allowed?

Answer: $f(g(x))$; x must be in the domain of g, and $g(x)$ must be in the domain of f.

1. Apply g first and then f.
2. Both stages must be defined.

Evidence anchors (0–3, descriptive only):

- 1 point: Apply g first and then f.

- 1 point: Both stages must be defined.
- 1 point: give f(g(x)); x must be in the domain of g, and g(x) must be in the domain of f. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Reversing composition; forgetting the outer function can add a restriction
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Compute the inner output first and ask whether the outer function accepts it.

Follow-up: If the reasoning is uncertain, use F1-R without revealing this solution. If independent reasoning is sound, consider F1-4. Do not infer a gap from an unreached item.

L1-1 · Implication and counterexamples · L1

Question: If a statement says “P implies Q”, what are its converse and contrapositive? Which is logically equivalent to the original?

Answer: Converse: Q implies P. Contrapositive: not Q implies not P; only the contrapositive is always equivalent.

1. An implication fails only when P is true and Q false.
2. The contrapositive has exactly that same failure case; the converse need not.

Evidence anchors (0–3, descriptive only):

- 1 point: An implication fails only when P is true and Q false.
- 1 point: The contrapositive has exactly that same failure case; the converse need not.
- 1 point: give Converse: Q implies P. Contrapositive: not Q implies not P; only the contrapositive is always equivalent. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

S1-1 · Arithmetic sequences and sums · L1

Question: State the nth term and sum of the first n terms of an arithmetic sequence with first term a and common difference d, for positive integer n.

Answer: $a_n=a+(n-1)d$; $S_n=n(2a+(n-1)d)/2$.

1. There are $n-1$ steps from the first term to the nth.
2. Pairing first/last terms gives the mean $(a+a_n)/2$ times n.

Evidence anchors (0–3, descriptive only):

- 1 point: There are n−1 steps from the first term to the nth.
- 1 point: Pairing first/last terms gives the mean $(a+a_n)/2$ times n.
- 1 point: give $a_n=a+(n-1)d$; $S_n=n(2a+(n-1)d)/2$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one term count; confusing last term with number of terms
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List the first few indices and compare the first and last terms.

Follow-up: If the reasoning is uncertain, use S1-R without revealing this solution. If independent reasoning is sound, consider S1-4. Do not infer a gap from an unreached item.

C1-1 · Cases, product rule and completeness · L1

Question: When should counts of possibilities be multiplied, and when should they be added? State the disjointness condition.

Answer: Multiply successive choices when each continuation count is correctly accounted for; add counts of disjoint cases.

1. A product counts complete choice sequences.
2. Addition without correction requires cases with no overlap; variable continuation counts must be summed appropriately.

Evidence anchors (0–3, descriptive only):

- 1 point: A product counts complete choice sequences.
- 1 point: Addition without correction requires cases with no overlap; variable continuation counts must be summed appropriately.
- 1 point: give Multiply successive choices when each continuation count is correctly accounted for; add counts of disjoint cases. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Allowing a leading zero; multiplying nonconstant continuation counts without splitting cases
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Specify exactly one completed object. Check whether every first choice has the same number of continuations.

Follow-up: If the reasoning is uncertain, use C1-R without revealing this solution. If independent reasoning is sound, consider C1-4. Do not infer a gap from an unreached item.

N3-1 · Last digits and power remainders · L1

Question: List the repeating last digits of positive powers of 7. Which cycle position corresponds to an exponent divisible by 4?

Answer: 7,9,3,1; a multiple of 4 uses the last position, digit 1.

1. Multiplying the current last digit by 7 produces the next last digit.
2. After four steps the digit is 1, so another multiplication restarts the cycle.
3. Remainder zero means position 4, not position 0.

Evidence anchors (0–3, descriptive only):

- 1 point: Multiplying the current last digit by 7 produces the next last digit.
- 1 point: After four steps the digit is 1, so another multiplication restarts the cycle.
- 1 point: give 7,9,3,1; a multiple of 4 uses the last position, digit 1. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

Day 2: Algebra and equations

Core: A1-3, A2-3, A3-3, A1-2, A2-2, A3-2, A2-1, A4-1. Reserve selection: A1-R, A1-4, A2-R, A2-4, A3-R, A3-4, A4-R, A4-4, EX-VJ.

A1-3 · Equivalent equations and domain · L3

Question: Solve over the reals: $(x^2 - 9)/(x - 3) = 6$. State all restrictions.

Answer: No solution; $x \neq 3$.

1. The denominator excludes x = 3.
2. For permitted x, cancel x − 3 to obtain x + 3 = 6.
3. The only candidate is 3, which is excluded.

Evidence anchors (0–3, descriptive only):

- 1 point: The denominator excludes x = 3.
- 1 point: For permitted x, cancel x − 3 to obtain x + 3 = 6.
- 1 point: give No solution; x ≠ 3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

A2-3 · Polynomial roots and standard theorems · L3

Question: Find all real roots of $2x^3 - 3x^2 - 8x + 12 = 0$.

Answer: −2, 3/2, 2.

1. A rational root must come from integer factors of 12 divided by factors of 2.
2. Factor by grouping: $x^2(2x - 3) - 4(2x - 3) = (2x - 3)(x - 2)(x + 2)$.
3. All three linear factors give the stated roots.

Evidence anchors (0–3, descriptive only):

- 1 point: A rational root must come from integer factors of 12 divided by factors of 2.
- 1 point: Factor by grouping: $x^2(2x - 3) - 4(2x - 3) = (2x - 3)(x - 2)(x + 2)$.
- 1 point: give −2, 3/2, 2. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating the rational-root list as a guarantee of roots; confusing P(a) with P(−a)
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: For division by x−a, think about substituting x=a. For rational roots, inspect the leading and constant coefficients.

Follow-up: If the reasoning is uncertain, use A2-R without revealing this solution. If independent reasoning is sound, consider A2-4. Do not infer a gap from an unreached item.

A3-3 · Vieta formulas and symmetric root expressions · L3

Question: The roots of $x^2 - 5x + 3$ are r and s. Find r^2+s^2 without calculating either root.

Answer: 19.

1. The sum is 5 and product is 3.
2. $r^2+s^2=(r+s)^2-2rs=25-6=19$.

Evidence anchors (0–3, descriptive only):

- 1 point: The sum is 5 and product is 3.
- 1 point: $r^2+s^2=(r+s)^2-2rs=25-6=19$.
- 1 point: give 19. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand (x−r)(x−s), or expand a square of a sum.

Follow-up: If the reasoning is uncertain, use A3-R without revealing this solution. If independent reasoning is sound, consider A3-4. Do not infer a gap from an unreached item.

A1-2 · Equivalent equations and domain · L2

Question: Solve over the reals: $x(x - 4) = 0$.

Answer: $x = 0$ or 4.

1. A product is zero only if at least one factor is zero.
2. Solve $x = 0$ and $x - 4 = 0$; both satisfy the original equation.

Evidence anchors (0–3, descriptive only):

- 1 point: A product is zero only if at least one factor is zero.
- 1 point: Solve $x = 0$ and $x - 4 = 0$; both satisfy the original equation.
- 1 point: give $x = 0$ or 4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

A2-2 · Polynomial roots and standard theorems · L2

Question: Find the remainder when $P(x)=2x^3 - 3x + 5$ is divided by $x - 2$.

Answer: 15.

1. The remainder is $P(2)$.
2. Compute $16 - 6 + 5 = 15$.

Evidence anchors (0–3, descriptive only):

- 1 point: The remainder is $P(2)$.
- 1 point: Compute $16 - 6 + 5 = 15$.
- 1 point: give 15. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating the rational-root list as a guarantee of roots; confusing $P(a)$ with $P(-a)$
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: For division by $x-a$, think about substituting $x=a$. For rational roots, inspect the leading and constant coefficients.

Follow-up: If the reasoning is uncertain, use A2-R without revealing this solution. If independent reasoning is sound, consider A2-4. Do not infer a gap from an unreached item.

A3-2 · Vieta formulas and symmetric root expressions · L2

Question: The roots of $x^2 - 7x + 10$ are r and s. Find $r+s$ and rs without solving for them.

Answer: 7 and 10.

1. Read the coefficients of the monic quadratic.
2. The negative x-coefficient gives the sum; the constant gives the product.

Evidence anchors (0–3, descriptive only):

- 1 point: Read the coefficients of the monic quadratic.
- 1 point: The negative x-coefficient gives the sum; the constant gives the product.
- 1 point: give 7 and 10. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand (x−r)(x−s), or expand a square of a sum.

Follow-up: If the reasoning is uncertain, use A3-R without revealing this solution. If independent reasoning is sound, consider A3-4. Do not infer a gap from an unreached item.

A2-1 · Polynomial roots and standard theorems · L1

Question: State the remainder theorem, factor theorem, and rational-root restriction for an integer-coefficient polynomial with nonzero leading coefficient.

Answer: Remainder on division by x − a is P(a); x − a divides P iff P(a)=0; a rational root p/q in lowest terms has p dividing the constant term and q dividing the leading coefficient.

1. Substituting a leaves the constant remainder.
2. Zero remainder is exactly divisibility by x − a.
3. For a reduced rational root, clearing denominators gives the divisibility restrictions. When the constant term is zero, first factor out x.

Evidence anchors (0–3, descriptive only):

- 1 point: state remainder P(a) for division by x−a.
- 1 point: state the factor iff zero-remainder equivalence.
- 1 point: give the reduced rational-root divisibility restrictions, including nonzero leading coefficient.

Possible error hypotheses, not automatic diagnoses:

- Treating the rational-root list as a guarantee of roots; confusing P(a) with P(−a)
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: For division by x−a, think about substituting x=a. For rational roots, inspect the leading and constant coefficients.

Follow-up: If the reasoning is uncertain, use A2-R without revealing this solution. If independent reasoning is sound, consider A2-4. Do not infer a gap from an unreached item.

A4-1 · Substitution and symmetry · L1

Question: For nonzero real x, expand $(x+1/x)^2$. What is x^2+1/x^2 in terms of $t=x+1/x$?

Answer: $(x+1/x)^2=x^2+2+1/x^2$; $x^2+1/x^2=t^2-2$.

1. The cross term is $2x(1/x)=2$.
2. Rearrange after setting $t=x+1/x$.

Evidence anchors (0–3, descriptive only):

- 1 point: The cross term is $2x(1/x)=2$.
- 1 point: Rearrange after setting $t=x+1/x$.
- 1 point: give $(x+1/x)^2=x^2+2+1/x^2$; $x^2+1/x^2=t^2-2$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating x^2 as x; losing the nonzero restriction or real feasibility of a substituted value
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Look for a repeated expression. If you replace it, record which real values the replacement can take.

Follow-up: If the reasoning is uncertain, use A4-R without revealing this solution. If independent reasoning is sound, consider A4-4. Do not infer a gap from an unreached item.

Day 3: Inequalities and bounds

Core: I1-3, I2-3, I3-3, I1-2, I2-2, I3-2, I2-1, I3-1. Reserve selection: I1-R, I1-4, I2-R, I2-4, I3-R, I3-4.

I1-3 · Signs and inequality transformations · L3

Question: Solve $(x-1)/(x+2)\geq 0$ over the reals.

Answer: $x<-2$ or $x\geq 1$.

1. Exclude $x=-2$ and mark the numerator zero at 1.
2. On the three intervals, signs are positive, negative, positive.
3. Include 1 because equality is allowed; exclude -2.

Evidence anchors (0–3, descriptive only):

- 1 point: Exclude $x=-2$ and mark the numerator zero at 1.
- 1 point: On the three intervals, signs are positive, negative, positive.
- 1 point: give $x<-2$ or $x\geq 1$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Failing to reverse order after negative division; multiplying by an expression of unknown sign
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Mark points where a factor is zero or undefined. Test one point in each interval.

Follow-up: If the reasoning is uncertain, use I1-R without revealing this solution. If independent reasoning is sound, consider I1-4. Do not infer a gap from an unreached item.

I2-3 · Absolute value and intervals · L3

Question: Find all real x satisfying $|x-1|+|x-5|=4$.

Answer: $1 \le x \le 5$.

1. Between 1 and 5, the two distances add to the endpoint separation 4.
2. To the left or right, the sum is larger than 4.
3. Include both endpoints.

Evidence anchors (0–3, descriptive only):

- 1 point: Between 1 and 5, the two distances add to the endpoint separation 4.
- 1 point: To the left or right, the sum is larger than 4.
- 1 point: give $1 \le x \le 5$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Dropping one side of the absolute-value condition; counting an interval as one solution
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Locate the stated centres on a number line and interpret absolute values as distances.

Follow-up: If the reasoning is uncertain, use I2-R without revealing this solution. If independent reasoning is sound, consider I2-4. Do not infer a gap from an unreached item.

I3-3 · Bounds and equality cases · L3

Question: Positive a,b have $a+b=10$. What is the greatest possible ab, and when is it attained?

Answer: 25, attained at $a=b=5$.

1. $(a-b)^2 \ge 0$ implies $(a+b)^2 \ge 4ab$.
2. Thus $ab \le 25$; equal values attain the bound.

Evidence anchors (0–3, descriptive only):

- 1 point: $(a-b)^2 \ge 0$ implies $(a+b)^2 \ge 4ab$.
- 1 point: Thus $ab \le 25$; equal values attain the bound.
- 1 point: give 25, attained at $a=b=5$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Applying AM–GM outside its domain; stating a bound without checking attainability
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Consider a nonnegative square. After a bound, check when equality can hold.

Follow-up: If the reasoning is uncertain, use I3-R without revealing this solution. If independent reasoning is sound, consider I3-4. Do not infer a gap from an unreached item.

I1-2 · Signs and inequality transformations · L2

Question: Solve $-3x+2<11$ over the reals.

Answer: $x>-3$.

1. Subtract 2: $-3x<9$.
2. Divide by -3 and reverse the sign.

Evidence anchors (0–3, descriptive only):

- 1 point: Subtract 2: $-3x<9$.
- 1 point: Divide by -3 and reverse the sign.
- 1 point: give $x>-3$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Failing to reverse order after negative division; multiplying by an expression of unknown sign
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Mark points where a factor is zero or undefined. Test one point in each interval.

Follow-up: If the reasoning is uncertain, use I1-R without revealing this solution. If independent reasoning is sound, consider I1-4. Do not infer a gap from an unreached item.

I2-2 · Absolute value and intervals · L2

Question: Solve $|2x-3|\le 5$ over the reals.

Answer: $-1\le x\le 4$.

1. Write $-5\le 2x-3\le 5$.
2. Add 3 and divide by 2.

Evidence anchors (0–3, descriptive only):

- 1 point: Write $-5\le 2x-3\le 5$.
- 1 point: Add 3 and divide by 2.
- 1 point: give $-1\le x\le 4$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Dropping one side of the absolute-value condition; counting an interval as one solution
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Locate the stated centres on a number line and interpret absolute values as distances.

Follow-up: If the reasoning is uncertain, use I2-R without revealing this solution. If independent reasoning is sound, consider I2-4. Do not infer a gap from an unreached item.

I3-2 · Bounds and equality cases · L2

Question: For $x>0$, find the minimum of $x+9/x$ and the x attaining it.

Answer: Minimum 6 at x=3.

1. The two positive terms have product 9.
2. AM–GM gives sum at least 6, with equality x=9/x, hence x=3.

Evidence anchors (0–3, descriptive only):

- 1 point: The two positive terms have product 9.
- 1 point: AM–GM gives sum at least 6, with equality x=9/x, hence x=3.
- 1 point: give Minimum 6 at x=3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Applying AM–GM outside its domain; stating a bound without checking attainability
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Consider a nonnegative square. After a bound, check when equality can hold.

Follow-up: If the reasoning is uncertain, use I3-R without revealing this solution. If independent reasoning is sound, consider I3-4. Do not infer a gap from an unreached item.

I2-1 · Absolute value and intervals · L1

Question: For a>0, express |x−c|≤a as an interval and explain it as distance.

Answer: c−a≤x≤c+a; x is at distance at most a from c.

1. Absolute difference is distance on the real line.
2. Both endpoints are allowed because the inequality is not strict.

Evidence anchors (0–3, descriptive only):

- 1 point: Absolute difference is distance on the real line.
- 1 point: Both endpoints are allowed because the inequality is not strict.
- 1 point: give c−a≤x≤c+a; x is at distance at most a from c. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Dropping one side of the absolute-value condition; counting an interval as one solution
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Locate the stated centres on a number line and interpret absolute values as distances.

Follow-up: If the reasoning is uncertain, use I2-R without revealing this solution. If independent reasoning is sound, consider I2-4. Do not infer a gap from an unreached item.

I3-1 · Bounds and equality cases · L1

Question: State the two-variable arithmetic–geometric mean inequality, its domain and equality condition.

Answer: For a,b≥0, (a+b)/2≥√(ab), with equality iff a=b.

1. The square roots require nonnegative a,b.
2. $(\sqrt{a}-\sqrt{b})^2 \geq 0$ proves $a+b \geq 2\sqrt{ab}$; equality requires a=b.

Evidence anchors (0–3, descriptive only):

- 1 point: The square roots require nonnegative a,b.
- 1 point: $(\sqrt{a}-\sqrt{b})^2 \geq 0$ proves $a+b \geq 2\sqrt{ab}$; equality requires a=b.
- 1 point: give For a,b≥0, $(a+b)/2 \geq \sqrt{ab}$, with equality iff a=b. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Applying AM–GM outside its domain; stating a bound without checking attainability
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Consider a nonnegative square. After a bound, check when equality can hold.

Follow-up: If the reasoning is uncertain, use I3-R without revealing this solution. If independent reasoning is sound, consider I3-4. Do not infer a gap from an unreached item.

Day 4: Functions and substitution

Core: F1-3, F2-3, A4-3, F1-2, F2-2, A4-2, F2-1, A5-1. Reserve selection: F1-R, F1-4, F2-R, F2-4, A4-R, A4-4, A5-R, A5-4.

F1-3 · Functions, domain and composition · L3

Question: Let $f(x)=\sqrt{x-1}$ and $g(x)=3-x$. Find the real domain of f(g(x)).

Answer: $x \leq 2$.

1. The square-root input becomes $2-x$.
2. It must be nonnegative, so $x \leq 2$.

Evidence anchors (0–3, descriptive only):

- 1 point: The square-root input becomes $2-x$.
- 1 point: It must be nonnegative, so $x \leq 2$.
- 1 point: give $x \leq 2$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Reversing composition; forgetting the outer function can add a restriction
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Compute the inner output first and ask whether the outer function accepts it.

Follow-up: If the reasoning is uncertain, use F1-R without revealing this solution. If independent reasoning is sound, consider F1-4. Do not infer a gap from an unreached item.

F2-3 · Reasoning from a functional equation · L3

Question: A function on R satisfies f(x)+2f(1−x)=3x for every real x. Find f(0).

Answer: 2.

1. At x=0, f(0)+2f(1)=0.
2. At x=1, f(1)+2f(0)=3.
3. Solving the pair gives f(0)=2, f(1)=−1.

Evidence anchors (0–3, descriptive only):

- 1 point: At x=0, f(0)+2f(1)=0.
- 1 point: At x=1, f(1)+2f(0)=3.
- 1 point: give 2. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Assuming an unproved formula for f; failing to exploit substituted arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try zero, a negative input or replacing an input with its partner. Each substitution must be allowed.

Follow-up: If the reasoning is uncertain, use F2-R without revealing this solution. If independent reasoning is sound, consider F2-4. Do not infer a gap from an unreached item.

A4-3 · Substitution and symmetry · L3

Question: Find all real x satisfying $x^4 - 3x^2 + 2=0$.

Answer: $x=\pm1, \pm\sqrt{2}$.

1. View the expression as a quadratic in x^2.
2. $(x^2-1)(x^2-2)=0$.
3. Both nonnegative values of x^2 give two real square roots.

Evidence anchors (0–3, descriptive only):

- 1 point: View the expression as a quadratic in x^2.
- 1 point: $(x^2-1)(x^2-2)=0$.
- 1 point: give $x=\pm1, \pm\sqrt{2}$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating x^2 as x; losing the nonzero restriction or real feasibility of a substituted value
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Look for a repeated expression. If you replace it, record which real values the replacement can take.

Follow-up: If the reasoning is uncertain, use A4-R without revealing this solution. If independent reasoning is sound, consider A4-4. Do not infer a gap from an unreached item.

F1-2 · Functions, domain and composition · L2

Question: Let $f(x)=x^2+1$ and $g(x)=2x-3$ for all real x. Find $f(g(2))$ and $g(f(2))$.

Answer: 2 and 7.

1. $g(2)=1$, so $f(g(2))=2$.
2. $f(2)=5$, so $g(f(2))=7$; composition order matters.

Evidence anchors (0–3, descriptive only):

- 1 point: $g(2)=1$, so $f(g(2))=2$.
- 1 point: $f(2)=5$, so $g(f(2))=7$; composition order matters.
- 1 point: give 2 and 7. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Reversing composition; forgetting the outer function can add a restriction
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Compute the inner output first and ask whether the outer function accepts it.

Follow-up: If the reasoning is uncertain, use F1-R without revealing this solution. If independent reasoning is sound, consider F1-4. Do not infer a gap from an unreached item.

F2-2 · Reasoning from a functional equation · L2

Question: For all real x,y, $f(x+y)=f(x)+f(y)$, and $f(1)=3$. Find $f(2)$, $f(-1)$ and $f(1/2)$.

Answer: 6, −3, 3/2.

1. Repeated addition gives $f(2)=6$.
2. Since $f(0)=0$, $f(-1)=-f(1)=-3$.
3. Two copies of $f(1/2)$ sum to $f(1)=3$.

Evidence anchors (0–3, descriptive only):

- 1 point: Repeated addition gives $f(2)=6$.
- 1 point: Since $f(0)=0$, $f(-1)=-f(1)=-3$.
- 1 point: give 6, −3, 3/2. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Assuming an unproved formula for f; failing to exploit substituted arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try zero, a negative input or replacing an input with its partner. Each substitution must be allowed.

Follow-up: If the reasoning is uncertain, use F2-R without revealing this solution. If independent reasoning is sound, consider F2-4. Do not infer a gap from an unreached item.

A4-2 · Substitution and symmetry · L2

Question: For $x \neq 0$, $x+1/x=3$. Find x^2+1/x^2.

Answer: 7.

1. Square both sides to get $x^2+2+1/x^2=9$.
2. Subtract 2.

Evidence anchors (0–3, descriptive only):

- 1 point: Square both sides to get $x^2+2+1/x^2=9$.
- 1 point: Subtract 2.
- 1 point: give 7. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating x^2 as x; losing the nonzero restriction or real feasibility of a substituted value
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Look for a repeated expression. If you replace it, record which real values the replacement can take.

Follow-up: If the reasoning is uncertain, use A4-R without revealing this solution. If independent reasoning is sound, consider A4-4. Do not infer a gap from an unreached item.

F2-1 · Reasoning from a functional equation · L1

Question: A real-valued function on R satisfies $f(x+y)=f(x)+f(y)$. What do the substitutions $x=y=0$ and $y=-x$ tell you about $f(0)$ and $f(-x)$?

Answer: $f(0)=0$ and $f(-x)=-f(x)$.

1. At $x=y=0$, $f(0)=2f(0)$, so $f(0)=0$.
2. At $y=-x$, $0=f(x)+f(-x)$, so $f(-x)=-f(x)$.

Evidence anchors (0–3, descriptive only):

- 1 point: make a valid zero substitution.
- 1 point: derive $f(0)=0$.
- 1 point: use $y=-x$ to derive $f(-x)=-f(x)$.

Possible error hypotheses, not automatic diagnoses:

- Assuming an unproved formula for f; failing to exploit substituted arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try zero, a negative input or replacing an input with its partner. Each substitution must be allowed.

Follow-up: If the reasoning is uncertain, use F2-R without revealing this solution. If independent reasoning is sound, consider F2-4. Do not infer a gap from an unreached item.

A5-1 · Exponents and logarithms · L1

Question: For real logarithms to base b, state the restrictions on b and on the argument. State log_b(uv) for positive u,v.

Answer: b>0, b≠1, argument>0; log_b(uv)=log_b u + log_b v.

1. The logarithm inverts exponentiation with a positive base other than 1.
2. Its real argument must be positive; the product rule follows from $b^{p+q}=b^p\,b^q$.

Evidence anchors (0–3, descriptive only):

- 1 point: The logarithm inverts exponentiation with a positive base other than 1.
- 1 point: Its real argument must be positive; the product rule follows from $b^{p+q}=b^p\,b^q$.
- 1 point: give b>0, b≠1, argument>0; log_b(uv)=log_b u + log_b v. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Combining unlike bases incorrectly; ignoring positive logarithm arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Express the quantities with one base, and record logarithm domains before transforming.

Follow-up: If the reasoning is uncertain, use A5-R without revealing this solution. If independent reasoning is sound, consider A5-4. Do not infer a gap from an unreached item.

Day 5: Logic and divisibility

Core: L1-3, L2-3, N1-3, L1-2, L2-2, N1-2, L2-1, N1-1. Reserve selection: L1-R, L1-4, L2-R, L2-4, N1-R, N1-4.

L1-3 · Implication and counterexamples · L3

Question: A student says: “n² is even, so n is even, because squaring an even number gives an even number.” Is the claim true, and is that argument enough? Repair it.

Answer: The claim is true, but that argument proves the opposite direction. If n were odd, n=2k+1 would give odd n²; hence even n² forces even n.

1. Separate the truth of the conclusion from the direction established.
2. Prove the contrapositive by expanding $(2k+1)^2=2(2k^2+2k)+1$.

Evidence anchors (0–3, descriptive only):

- 1 point: Separate the truth of the conclusion from the direction established.
- 1 point: Prove the contrapositive by expanding $(2k+1)^2=2(2k^2+2k)+1$.
- 1 point: give The claim is true, but that argument proves the opposite direction. If n were odd, n=2k+1 would give odd n²; hence even n² forces even n. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

L2-3 · Quantifiers and negation · L3

Question: Over the integers, compare: (A) for every x there exists y with y>x; (B) there exists y such that for every x, y>x. Which are true?

Answer: A true; B false.

1. For A choose y=x+1 separately for each x.
2. For B a fixed y fails when x=y, or when x=y+1.

Evidence anchors (0–3, descriptive only):

- 1 point: For A choose y=x+1 separately for each x.
- 1 point: For B a fixed y fails when x=y, or when x=y+1.
- 1 point: give A true; B false. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Swapping quantifier order; using one fixed witness when it must depend on an input
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the quantifiers in order and identify which choices may depend on earlier choices.

Follow-up: If the reasoning is uncertain, use L2-R without revealing this solution. If independent reasoning is sound, consider L2-4. Do not infer a gap from an unreached item.

N1-3 · Divisibility from place value · L3

Question: Explain why a decimal integer is divisible by 11 iff its alternating digit sum is divisible by 11.

Answer: Each power of 10 has the same remainder as the corresponding power of −1 modulo 11.

1. Write the place-value expansion from the units digit.
2. Since 10=11−1, powers have alternating remainders 1,−1.
3. Their weighted sum differs from the alternating digit sum by a multiple of 11.

Evidence anchors (0–3, descriptive only):

- 1 point: Write the place-value expansion from the units digit.
- 1 point: Since 10=11−1, powers have alternating remainders 1,−1.
- 1 point: give Each power of 10 has the same remainder as the corresponding power of −1 modulo 11. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Remembering a digit trick without its base-dependent explanation

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number as a sum of digits times powers of its base.

Follow-up: If the reasoning is uncertain, use N1-R without revealing this solution. If independent reasoning is sound, consider N1-4. Do not infer a gap from an unreached item.

L1-2 · Implication and counterexamples · L2

Question: For integers n, assess both claims: if 4 divides n then 2 divides n; if 2 divides n then 4 divides n.

Answer: First true; second false, for example n=2.

1. Write n=4k=2(2k) to prove the first.
2. The integer 2 satisfies the premise but not the conclusion of the second.

Evidence anchors (0–3, descriptive only):

- 1 point: Write n=4k=2(2k) to prove the first.
- 1 point: The integer 2 satisfies the premise but not the conclusion of the second.
- 1 point: give First true; second false, for example n=2. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

L2-2 · Quantifiers and negation · L2

Question: Negate precisely: “For every integer n there exists an integer m with m>n.” Is the original statement true?

Answer: Negation: there exists integer n such that every integer m satisfies m≤n. The original is true, since m=n+1 works.

1. Reverse each quantifier in order and negate m>n to m≤n.
2. A witness for the original may depend on n.

Evidence anchors (0–3, descriptive only):

- 1 point: Reverse each quantifier in order and negate m>n to m≤n.
- 1 point: A witness for the original may depend on n.
- 1 point: give Negation: there exists integer n such that every integer m satisfies m≤n. The original is true, since m=n+1 works. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Swapping quantifier order; using one fixed witness when it must depend on an input
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the quantifiers in order and identify which choices may depend on earlier choices.

Follow-up: If the reasoning is uncertain, use L2-R without revealing this solution. If independent reasoning is sound, consider L2-4. Do not infer a gap from an unreached item.

N1-2 · Divisibility from place value · L2

Question: For which digit d is the decimal number 4d2 divisible by 9?

Answer: d=3.

1. The digit sum is 6+d.
2. For d between 0 and 9, its only possible multiple of 9 is 9, so d=3.

Evidence anchors (0–3, descriptive only):

- 1 point: The digit sum is 6+d.
- 1 point: For d between 0 and 9, its only possible multiple of 9 is 9, so d=3.
- 1 point: give d=3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Remembering a digit trick without its base-dependent explanation
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number as a sum of digits times powers of its base.

Follow-up: If the reasoning is uncertain, use N1-R without revealing this solution. If independent reasoning is sound, consider N1-4. Do not infer a gap from an unreached item.

L2-1 · Quantifiers and negation · L1

Question: Negate: “For every real x, P(x) holds.” Negate: “There exists a real x for which P(x) holds.”

Answer: There exists real x for which P(x) fails; for every real x, P(x) fails.

1. Negation exchanges every and exists.
2. It also negates the property, while retaining the same domain.

Evidence anchors (0–3, descriptive only):

- 1 point: Negation exchanges every and exists.
- 1 point: It also negates the property, while retaining the same domain.
- 1 point: give There exists real x for which P(x) fails; for every real x, P(x) fails. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Swapping quantifier order; using one fixed witness when it must depend on an input

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the quantifiers in order and identify which choices may depend on earlier choices.

Follow-up: If the reasoning is uncertain, use L2-R without revealing this solution. If independent reasoning is sound, consider L2-4. Do not infer a gap from an unreached item.

N1-1 · Divisibility from place value · L1

Question: State the tests for divisibility by 3 and 9 for a nonnegative integer in decimal notation.

Answer: The integer is divisible by 3 or 9 respectively iff its digit sum is divisible by that divisor.

1. A decimal number is a sum of digits times powers of 10.
2. Since 10 leaves remainder 1 on division by either 3 or 9, the number and digit sum have the same remainder.

Evidence anchors (0–3, descriptive only):

- 1 point: A decimal number is a sum of digits times powers of 10.
- 1 point: Since 10 leaves remainder 1 on division by either 3 or 9, the number and digit sum have the same remainder.
- 1 point: give The integer is divisible by 3 or 9 respectively iff its digit sum is divisible by that divisor. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Remembering a digit trick without its base-dependent explanation
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number as a sum of digits times powers of its base.

Follow-up: If the reasoning is uncertain, use N1-R without revealing this solution. If independent reasoning is sound, consider N1-4. Do not infer a gap from an unreached item.

Day 6: Sequences and sums

Core: S1-3, S2-3, S3-3, S1-2, S2-2, S3-2, S2-1, S3-1. Reserve selection: S1-R, S1-4, S2-R, S2-4, S3-R, S3-4.

S1-3 · Arithmetic sequences and sums · L3

Question: A theatre has 20 rows. The first has 12 seats and each later row has 2 more than the previous row. How many seats are there?

Answer: 620.

1. The last row has $12+19\cdot 2=50$ seats.
2. Twenty rows times the average 31 gives 620.

Evidence anchors (0–3, descriptive only):

- 1 point: The last row has 12+19·2=50 seats.
- 1 point: Twenty rows times the average 31 gives 620.
- 1 point: give 620. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one term count; confusing last term with number of terms
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List the first few indices and compare the first and last terms.

Follow-up: If the reasoning is uncertain, use S1-R without revealing this solution. If independent reasoning is sound, consider S1-4. Do not infer a gap from an unreached item.

S2-3 · Geometric sequences and sums · L3

Question: Write 0.272727… as a fraction in lowest terms.

Answer: 3/11.

1. Let u be the decimal; multiplying by 100 shifts one repeating block.
2. Subtract u from 100u to get 99u=27, so u=27/99=3/11.

Evidence anchors (0–3, descriptive only):

- 1 point: Let u be the decimal; multiplying by 100 shifts one repeating block.
- 1 point: Subtract u from 100u to get 99u=27, so u=27/99=3/11.
- 1 point: give 3/11. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one exponent; dividing by r−1 when r=1; using a divergent infinite sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number of terms. Compare a finite sum with the same sum multiplied by its ratio.

Follow-up: If the reasoning is uncertain, use S2-R without revealing this solution. If independent reasoning is sound, consider S2-4. Do not infer a gap from an unreached item.

S3-3 · Integer, square and cube sums · L3

Question: Find Σ k(k+1), for k=1,2,…,10.

Answer: 440.

1. Expand $k(k+1)=k^2+k$.
2. Sum squares and integers separately: 385+55=440.

Evidence anchors (0–3, descriptive only):

- 1 point: Expand $k(k+1)=k^2+k$.
- 1 point: Sum squares and integers separately: 385+55=440.

• 1 point: give 440. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

S1-2 · Arithmetic sequences and sums · L2

Question: Find $3+7+11+\cdots+99$.

Answer: 1275.

1. Solve $3+4(n-1)=99$ to get $n=25$.
2. The sum is $25(3+99)/2=1275$.

Evidence anchors (0–3, descriptive only):

• 1 point: Solve $3+4(n-1)=99$ to get $n=25$.
• 1 point: The sum is $25(3+99)/2=1275$.
• 1 point: give 1275. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Off-by-one term count; confusing last term with number of terms
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List the first few indices and compare the first and last terms.

Follow-up: If the reasoning is uncertain, use S1-R without revealing this solution. If independent reasoning is sound, consider S1-4. Do not infer a gap from an unreached item.

S2-2 · Geometric sequences and sums · L2

Question: Find $1+3+3^2+\cdots+3^7$.

Answer: 3280.

1. There are 8 terms.
2. The geometric formula gives $(3^8-1)/(3-1)=6560/2=3280$.

Evidence anchors (0–3, descriptive only):

• 1 point: There are 8 terms.
• 1 point: The geometric formula gives $(3^8-1)/(3-1)=6560/2=3280$.
• 1 point: give 3280. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one exponent; dividing by r−1 when r=1; using a divergent infinite sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number of terms. Compare a finite sum with the same sum multiplied by its ratio.

Follow-up: If the reasoning is uncertain, use S2-R without revealing this solution. If independent reasoning is sound, consider S2-4. Do not infer a gap from an unreached item.

S3-2 · Integer, square and cube sums · L2

Question: Calculate $1^2+2^2+\cdots+10^2$.

Answer: 385.

1. Apply the square-sum formula with n=10.
2. Compute 10·11·21/6=385.

Evidence anchors (0–3, descriptive only):

- 1 point: Apply the square-sum formula with n=10.
- 1 point: Compute 10·11·21/6=385.
- 1 point: give 385. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

S2-1 · Geometric sequences and sums · L1

Question: State the finite geometric sum $1+r+\cdots+r^{n-1}$ for positive integer n. Include r=1. When does the corresponding infinite series converge?

Answer: $(1-r^n)/(1-r)$ if $r\neq1$; n if r=1. The infinite series converges for $|r|<1$, to $1/(1-r)$.

1. Multiplying the finite sum by r and subtracting cancels the interior terms.
2. For the infinite limit, r^n tends to zero exactly when $|r|<1$.

Evidence anchors (0–3, descriptive only):

- 1 point: Multiplying the finite sum by r and subtracting cancels the interior terms.
- 1 point: For the infinite limit, r^n tends to zero exactly when $|r|<1$.
- 1 point: give $(1-r^n)/(1-r)$ if $r\neq1$; n if r=1. The infinite series converges for $|r|<1$, to $1/(1-r)$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one exponent; dividing by r−1 when r=1; using a divergent infinite sum

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number of terms. Compare a finite sum with the same sum multiplied by its ratio.

Follow-up: If the reasoning is uncertain, use S2-R without revealing this solution. If independent reasoning is sound, consider S2-4. Do not infer a gap from an unreached item.

S3-1 · Integer, square and cube sums · L1

Question: State formulas for Σk, Σk^2 and Σk^3, where k runs from 1 to positive integer n.

Answer: $n(n+1)/2$; $n(n+1)(2n+1)/6$; $[n(n+1)/2]^2$.

1. State each formula with the same lower and upper limits.
2. Check n=1 to detect missing factors; knowing one does not imply knowing the others.

Evidence anchors (0–3, descriptive only):

- 1 point: correct integer-sum formula and limits.
- 1 point: correct square-sum formula and limits.
- 1 point: correct cube-sum formula and limits.

Possible error hypotheses, not automatic diagnoses:

- Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

Day 7: Counting without duplication

Core: C1-3, C2-3, C3-3, C1-2, C2-2, C3-2, C2-1, C3-1. Reserve selection: C1-R, C1-4, C2-R, C2-4, C3-R, C3-4.

C1-3 · Cases, product rule and completeness · L3

Question: How many three-digit positive integers have an even last digit and no repeated digits?

Answer: 328.

1. Split by last digit zero or nonzero even.
2. Last digit 0 gives 9·8=72. Last digit 2,4,6,8 gives 4·8·8=256.
3. Add the disjoint cases: 328.

Evidence anchors (0–3, descriptive only):

- 1 point: Split by last digit zero or nonzero even.
- 1 point: Last digit 0 gives 9·8=72. Last digit 2,4,6,8 gives 4·8·8=256.

• 1 point: give 328. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Allowing a leading zero; multiplying nonconstant continuation counts without splitting cases
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Specify exactly one completed object. Check whether every first choice has the same number of continuations.

Follow-up: If the reasoning is uncertain, use C1-R without revealing this solution. If independent reasoning is sound, consider C1-4. Do not infer a gap from an unreached item.

C2-3 · Choosing, ordering and complements · L3

Question: A committee of 3 is chosen from 5 women and 4 men. How many committees contain at least one woman?

Answer: 80.

1. There are C(9,3)=84 committees in all.
2. Exclude the C(4,3)=4 all-male committees to get 80.

Evidence anchors (0–3, descriptive only):

• 1 point: There are C(9,3)=84 committees in all.
• 1 point: Exclude the C(4,3)=4 all-male committees to get 80.
• 1 point: give 80. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Counting an unordered committee as ordered; multiplying by a role count twice
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

C3-3 · Overlapping cases and inclusion–exclusion · L3

Question: How many integers from 1 through 100 are divisible by 3 or 5?

Answer: 47.

1. Multiples of 3: 33; multiples of 5: 20.
2. Their overlap is the 6 multiples of 15.
3. Add 33+20−6=47.

Evidence anchors (0–3, descriptive only):

• 1 point: Multiples of 3: 33; multiples of 5: 20.
• 1 point: Their overlap is the 6 multiples of 15.

- 1 point: give 47. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Double-counting intersections; subtracting a triple overlap too often
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Place one object belonging to every relevant set into your formula and count its net contribution.

Follow-up: If the reasoning is uncertain, use C3-R without revealing this solution. If independent reasoning is sound, consider C3-4. Do not infer a gap from an unreached item.

C1-2 · Cases, product rule and completeness · L2

Question: A code has one letter from A,B,C followed by two digits from 0,…,9. Repetition is allowed. How many codes exist?

Answer: 300.

1. There are 3 choices for the letter and 10 for each digit.
2. Multiply 3·10·10.

Evidence anchors (0–3, descriptive only):

- 1 point: There are 3 choices for the letter and 10 for each digit.
- 1 point: Multiply 3·10·10.
- 1 point: give 300. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Allowing a leading zero; multiplying nonconstant continuation counts without splitting cases
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Specify exactly one completed object. Check whether every first choice has the same number of continuations.

Follow-up: If the reasoning is uncertain, use C1-R without revealing this solution. If independent reasoning is sound, consider C1-4. Do not infer a gap from an unreached item.

C2-2 · Choosing, ordering and complements · L2

Question: From 8 distinct students, how many groups of 3 can be chosen? How many ways can president, secretary and treasurer be assigned to different students?

Answer: 56 groups; 336 assignments.

1. The group has no order, giving C(8,3)=56.
2. The offices are distinct, giving 8·7·6=336.

Evidence anchors (0–3, descriptive only):

- 1 point: The group has no order, giving C(8,3)=56.
- 1 point: The offices are distinct, giving 8·7·6=336.
- 1 point: give 56 groups; 336 assignments. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Counting an unordered committee as ordered; multiplying by a role count twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

C3-2 · Overlapping cases and inclusion–exclusion · L2

Question: In a class of 30, 18 study French and 15 study German; 7 study both. How many study at least one, and how many neither?

Answer: 26 and 4.

1. The union size is $18+15-7=26$.
2. Subtract from the class size to get 4 studying neither.

Evidence anchors (0–3, descriptive only):

- 1 point: The union size is $18+15-7=26$.
- 1 point: Subtract from the class size to get 4 studying neither.
- 1 point: give 26 and 4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Double-counting intersections; subtracting a triple overlap too often
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Place one object belonging to every relevant set into your formula and count its net contribution.

Follow-up: If the reasoning is uncertain, use C3-R without revealing this solution. If independent reasoning is sound, consider C3-4. Do not infer a gap from an unreached item.

C2-1 · Choosing, ordering and complements · L1

Question: Explain the difference between choosing r people from n and assigning r distinct ordered positions to n people. State both counts for $0\le r\le n$.

Answer: Unordered choice: $C(n,r)=n!/[r!(n-r)!]$. Ordered assignment: $n!/(n-r)!$.

1. Each chosen set has r! orders.
2. Multiplying the combination count by r! gives the ordered count; assume distinct people and no repeated assignment.

Evidence anchors (0–3, descriptive only):

- 1 point: Each chosen set has r! orders.
- 1 point: Multiplying the combination count by r! gives the ordered count; assume distinct people and no repeated assignment.

- 1 point: give Unordered choice: C(n,r)=n!/[r!(n−r)!]. Ordered assignment: n!/(n−r)!. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Counting an unordered committee as ordered; multiplying by a role count twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

C3-1 · Overlapping cases and inclusion–exclusion · L1

Question: State the formula for the size of the union of two finite sets A and B. Why is there a subtraction?

Answer: |A□B|=|A|+|B|−|A∩B|.

1. Elements in the intersection were counted twice in the separate totals.
2. Subtract one copy to count each union element once.

Evidence anchors (0–3, descriptive only):

- 1 point: Elements in the intersection were counted twice in the separate totals.
- 1 point: Subtract one copy to count each union element once.
- 1 point: give |A∪B|=|A|+|B|−|A∩B|. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Double-counting intersections; subtracting a triple overlap too often
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Place one object belonging to every relevant set into your formula and count its net contribution.

Follow-up: If the reasoning is uncertain, use C3-R without revealing this solution. If independent reasoning is sound, consider C3-4. Do not infer a gap from an unreached item.

Day 8: Integer constraints and differences

Core: N2-3, N3-3, S4-3, N2-2, N3-2, S4-2, N2-1, S4-1. Reserve selection: N2-R, N2-4, N3-R, N3-4, S4-R, S4-4, N3-X1, N3-X2.

N2-3 · Gcd and integer constraints · L3

Question: Can 84a+30b=5 hold for integers a,b? Explain.

Answer: No.

1. Every integer combination of 84 and 30 is divisible by their gcd 6.
2. Five is not divisible by 6, so the equation is impossible.

Evidence anchors (0–3, descriptive only):

- 1 point: Every integer combination of 84 and 30 is divisible by their gcd 6.
- 1 point: Five is not divisible by 6, so the equation is impossible.
- 1 point: give No. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Solving over reals instead of integers; forgetting coprimality after dividing by the gcd
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Find a divisor common to both coefficients. When scaling by a gcd, state the new pair must be coprime.

Follow-up: If the reasoning is uncertain, use N2-R without revealing this solution. If independent reasoning is sound, consider N2-4. Do not infer a gap from an unreached item.

N3-3 · Last digits and power remainders · L3

Question: Find the remainder when 3^{100} is divided by 7. Show a short cycle or repeated-power argument.

Answer: 4.

1. Powers of 3 modulo 7 cycle through 3,2,6,4,5,1.
2. Since 100 leaves remainder 4 on division by 6, use the fourth value, 4.

Evidence anchors (0–3, descriptive only):

- 1 point: Powers of 3 modulo 7 cycle through 3,2,6,4,5,1.
- 1 point: Since 100 leaves remainder 4 on division by 6, use the fourth value, 4.
- 1 point: give 4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

S4-3 · Differences and telescoping · L3

Question: Find $\Sigma\ 1/[k(k+1)]$ for $k=1,\ldots,20$.

Answer: 20/21.

1. Use $1/[k(k+1)]=1/k-1/(k+1)$.
2. The sum telescopes to $1-1/21$.

Evidence anchors (0–3, descriptive only):

- 1 point: Use $1/[k(k+1)]=1/k-1/(k+1)$.

- 1 point: The sum telescopes to 1−1/21.
- 1 point: give 20/21. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling endpoints incorrectly; overlooking a shifted index
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the first three terms and last two terms, keeping their signs.

Follow-up: If the reasoning is uncertain, use S4-R without revealing this solution. If independent reasoning is sound, consider S4-4. Do not infer a gap from an unreached item.

N2-2 · Gcd and integer constraints · L2

Question: Find gcd(252,198), showing divisions.

Answer: 18.

1. 252=198+54, 198=3·54+36.
2. Then 54=36+18 and 36=2·18, so the last nonzero remainder is 18.

Evidence anchors (0–3, descriptive only):

- 1 point: 252=198+54, 198=3·54+36.
- 1 point: Then 54=36+18 and 36=2·18, so the last nonzero remainder is 18.
- 1 point: give 18. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Solving over reals instead of integers; forgetting coprimality after dividing by the gcd
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Find a divisor common to both coefficients. When scaling by a gcd, state the new pair must be coprime.

Follow-up: If the reasoning is uncertain, use N2-R without revealing this solution. If independent reasoning is sound, consider N2-4. Do not infer a gap from an unreached item.

N3-2 · Last digits and power remainders · L2

Question: Find the last digit of 7^{2026}.

Answer: 9.

1. The cycle length is 4 and 2026 leaves remainder 2 on division by 4.
2. The second last digit in the cycle is 9.

Evidence anchors (0–3, descriptive only):

- 1 point: The cycle length is 4 and 2026 leaves remainder 2 on division by 4.
- 1 point: The second last digit in the cycle is 9.
- 1 point: give 9. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

S4-2 · Differences and telescoping · L2

Question: Find Σ[1/k−1/(k+1)] for k=1,…,9.

Answer: 9/10.

1. The intermediate reciprocals cancel.
2. Only 1−1/10 remains.

Evidence anchors (0–3, descriptive only):

- 1 point: The intermediate reciprocals cancel.
- 1 point: Only 1−1/10 remains.
- 1 point: give 9/10. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling endpoints incorrectly; overlooking a shifted index
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the first three terms and last two terms, keeping their signs.

Follow-up: If the reasoning is uncertain, use S4-R without revealing this solution. If independent reasoning is sound, consider S4-4. Do not infer a gap from an unreached item.

N2-1 · Gcd and integer constraints · L1

Question: Explain why gcd(a,b)=gcd(b,a−qb) for integers a,b,q, not both a,b zero.

Answer: The two pairs have exactly the same common divisors.

1. A divisor of a and b also divides a−qb.
2. A divisor of b and a−qb also divides a=(a−qb)+qb.
3. Thus the positive greatest common divisor is unchanged.

Evidence anchors (0–3, descriptive only):

- 1 point: A divisor of a and b also divides a−qb.
- 1 point: A divisor of b and a−qb also divides a=(a−qb)+qb.
- 1 point: give The two pairs have exactly the same common divisors. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Solving over reals instead of integers; forgetting coprimality after dividing by the gcd
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Find a divisor common to both coefficients. When scaling by a gcd, state the new pair must be coprime.

Follow-up: If the reasoning is uncertain, use N2-R without revealing this solution. If independent reasoning is sound, consider N2-4. Do not infer a gap from an unreached item.

S4-1 · Differences and telescoping · L1

Question: If $a_k = b_{k+1} - b_k$, what is Σa_k from k=1 to n? Explain which terms remain.

Answer: $b_{n+1} - b_1$.

1. Write consecutive differences in order.
2. Every interior b term appears once positively and once negatively, leaving the endpoints.

Evidence anchors (0–3, descriptive only):

- 1 point: Write consecutive differences in order.
- 1 point: Every interior b term appears once positively and once negatively, leaving the endpoints.
- 1 point: give $b_{n+1} - b_1$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling endpoints incorrectly; overlooking a shifted index
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the first three terms and last two terms, keeping their signs.

Follow-up: If the reasoning is uncertain, use S4-R without revealing this solution. If independent reasoning is sound, consider S4-4. Do not infer a gap from an unreached item.

Day 9: Combining representations

Core: A5-3, S5-3, C4-3, A5-2, S5-2, C4-2, S5-1, C4-1. Reserve selection: A5-R, A5-4, S5-R, S5-4, C4-R, C4-4.

A5-3 · Exponents and logarithms · L3

Question: Solve $4^x - 5 \cdot 2^x + 4 = 0$ over the reals.

Answer: x=0 or 2.

1. Set $y = 2^x > 0$; then $4^x = y^2$.
2. $y^2 - 5y + 4 = (y-1)(y-4)$, so y=1 or 4.
3. These correspond to x=0 or 2.

Evidence anchors (0–3, descriptive only):

- 1 point: Set $y=2^x>0$; then $4^x=y^2$.
- 1 point: $y^2-5y+4=(y-1)(y-4)$, so $y=1$ or 4.
- 1 point: give $x=0$ or 2. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Combining unlike bases incorrectly; ignoring positive logarithm arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Express the quantities with one base, and record logarithm domains before transforming.

Follow-up: If the reasoning is uncertain, use A5-R without revealing this solution. If independent reasoning is sound, consider A5-4. Do not infer a gap from an unreached item.

S5-3 · Arithmetic–geometric sums · L3

Question: On day k a machine produces $k\cdot 2^k$ parts. Find the total over days 1 through 6 without listing every individual part.

Answer: 642.

1. Let $T=\Sigma k2^k$.
2. Subtract T from 2T after aligning powers: $T=6\cdot 2^7-(2+4+\cdots+64)$.
3. The geometric sum is 126, so $T=768-126=642$.

Evidence anchors (0–3, descriptive only):

- 1 point: Let $T=\Sigma k2^k$.
- 1 point: Subtract T from 2T after aligning powers: $T=6\cdot 2^7-(2+4+\cdots+64)$.
- 1 point: give 642. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating weighted terms as geometric; misaligning coefficients after shifting
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the sum and a ratio-multiple underneath it, aligning equal powers.

Follow-up: If the reasoning is uncertain, use S5-R without revealing this solution. If independent reasoning is sound, consider S5-4. Do not infer a gap from an unreached item.

C4-3 · Bijections and counting two ways · L3

Question: Ten people each shake hands with every other person once. How many handshakes occur? Explain why counting 10·9 needs correction.

Answer: 45.

1. Ten people each have nine partners, but this counts each pair twice.
2. Divide 90 by 2 to obtain 45 distinct handshakes.

Evidence anchors (0–3, descriptive only):

- 1 point: Ten people each have nine partners, but this counts each pair twice.
- 1 point: Divide 90 by 2 to obtain 45 distinct handshakes.
- 1 point: give 45. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Missing the inverse correspondence; counting each unordered pair twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Define what one object on each side represents, and explain how to recover one from the other.

Follow-up: If the reasoning is uncertain, use C4-R without revealing this solution. If independent reasoning is sound, consider C4-4. Do not infer a gap from an unreached item.

A5-2 · Exponents and logarithms · L2

Question: Solve $2^{x+1}=16$ over the reals.

Answer: x=3.

1. Write $16=2^4$.
2. Equal powers of base 2 have equal exponents, so $x+1=4$.

Evidence anchors (0–3, descriptive only):

- 1 point: Write $16=2^4$.
- 1 point: Equal powers of base 2 have equal exponents, so $x+1=4$.
- 1 point: give x=3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Combining unlike bases incorrectly; ignoring positive logarithm arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Express the quantities with one base, and record logarithm domains before transforming.

Follow-up: If the reasoning is uncertain, use A5-R without revealing this solution. If independent reasoning is sound, consider A5-4. Do not infer a gap from an unreached item.

S5-2 · Arithmetic–geometric sums · L2

Question: Find $1\cdot2+2\cdot2^2+3\cdot2^3+4\cdot2^4$.

Answer: 98.

1. List the four terms as 2,8,24,64.
2. Their sum is 98; a shift-and-subtract method is also valid.

Evidence anchors (0–3, descriptive only):

- 1 point: List the four terms as 2,8,24,64.
- 1 point: Their sum is 98; a shift-and-subtract method is also valid.

• 1 point: give 98. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Treating weighted terms as geometric; misaligning coefficients after shifting
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the sum and a ratio-multiple underneath it, aligning equal powers.

Follow-up: If the reasoning is uncertain, use S5-R without revealing this solution. If independent reasoning is sound, consider S5-4. Do not infer a gap from an unreached item.

C4-2 · Bijections and counting two ways · L2

Question: How many shortest grid paths go from (0,0) to (3,2), using only one-unit right and up moves? Explain a correspondence.

Answer: 10.

1. Each path corresponds to a length-5 string containing three R and two U symbols.
2. Choose the two U positions in C(5,2)=10 ways; the string determines one path.

Evidence anchors (0–3, descriptive only):

• 1 point: Each path corresponds to a length-5 string containing three R and two U symbols.
• 1 point: Choose the two U positions in C(5,2)=10 ways; the string determines one path.
• 1 point: give 10. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

• Missing the inverse correspondence; counting each unordered pair twice
• A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Define what one object on each side represents, and explain how to recover one from the other.

Follow-up: If the reasoning is uncertain, use C4-R without revealing this solution. If independent reasoning is sound, consider C4-4. Do not infer a gap from an unreached item.

S5-1 · Arithmetic–geometric sums · L1

Question: Let $T=1\cdot r+2\cdot r^2+\cdots+n\cdot r^n$. Write $T-rT$ with like powers grouped. Do not assume $r\neq 1$.

Answer: $T-rT=r+r^2+\square+r^n-n\,r^{n+1}$.

1. In rT, the coefficient of r^k for $2\le k\le n$ is $k-1$.
2. Subtraction leaves coefficient 1 at each $r,\ldots,r^n$ and the final term $-n\,r^{n+1}$. This identity also holds at $r=1$.

Evidence anchors (0–3, descriptive only):

• 1 point: In rT, the coefficient of r^k for $2\le k\le n$ is $k-1$.
• 1 point: Subtraction leaves coefficient 1 at each $r,\ldots,r^n$ and the final term $-n\,r^{n+1}$. This identity also holds at $r=1$.
• 1 point: give $T-rT=r+r^2+\cdots+r^n-n\,r^{n+1}$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating weighted terms as geometric; misaligning coefficients after shifting
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the sum and a ratio-multiple underneath it, aligning equal powers.

Follow-up: If the reasoning is uncertain, use S5-R without revealing this solution. If independent reasoning is sound, consider S5-4. Do not infer a gap from an unreached item.

C4-1 · Bijections and counting two ways · L1

Question: What must you show to count one finite collection by matching it to another?

Answer: A bijection: every object maps to exactly one object, every target occurs, and no two sources share a target; the map has an inverse.

1. Define the correspondence rather than only comparing examples.
2. Establish both uniqueness and exhaustiveness, for example by giving its inverse.

Evidence anchors (0–3, descriptive only):

- 1 point: Define the correspondence rather than only comparing examples.
- 1 point: Establish both uniqueness and exhaustiveness, for example by giving its inverse.
- 1 point: give A bijection: every object maps to exactly one object, every target occurs, and no two sources share a target; the map has an inverse. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Missing the inverse correspondence; counting each unordered pair twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Define what one object on each side represents, and explain how to recover one from the other.

Follow-up: If the reasoning is uncertain, use C4-R without revealing this solution. If independent reasoning is sound, consider C4-4. Do not infer a gap from an unreached item.

Day 10: Guarantees and delayed confirmation

Core: C5-3, D10-A1, D10-S3, D10-C2, D10-N3, D10-L1, C5-2, C5-1. Reserve selection: C5-R, C5-4, A1-R, A1-4, S3-R, S3-4, C2-R, C2-4, N3-R, N3-4, L1-R, L1-4.

C5-3 · Pigeonhole reasoning · L3

Question: Five integers are chosen. Show that two differ by a multiple of 4.

Answer: Two have the same remainder on division by 4, so their difference is divisible by 4.

1. There are four remainder classes 0,1,2,3.
2. Five integers force two into the same class.
3. Subtract their expressions 4a+r and 4b+r.

Evidence anchors (0–3, descriptive only):

- 1 point: There are four remainder classes 0,1,2,3.
- 1 point: Five integers force two into the same class.
- 1 point: give Two have the same remainder on division by 4, so their difference is divisible by 4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Claiming a likely event is guaranteed; choosing boxes that do not force the desired pair
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Choose categories so that two objects in one category have the required property.

Follow-up: If the reasoning is uncertain, use C5-R without revealing this solution. If independent reasoning is sound, consider C5-4. Do not infer a gap from an unreached item.

D10-A1 · Equivalent equations and domain · L3

Question: Solve over the reals: $(x^2-16)/(x-4)=8$. State restrictions.

Answer: No solution; $x\neq4$.

1. The original denominator excludes 4.
2. Cancellation gives $x+4=8$; its only candidate 4 is excluded.

Evidence anchors (0–3, descriptive only):

- 1 point: The original denominator excludes 4.
- 1 point: Cancellation gives $x+4=8$; its only candidate 4 is excluded.
- 1 point: conclude No solution; $x\neq4$. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

D10-S3 · Integer, square and cube sums · L3

Question: Find $\Sigma\, k(k-1)$, $k=1,\ldots,6$.

Answer: 70.

1. Expand to k^2-k.
2. The square sum is 91 and integer sum is 21; subtract.

Evidence anchors (0–3, descriptive only):

- 1 point: Expand to k^2-k.
- 1 point: The square sum is 91 and integer sum is 21; subtract.
- 1 point: conclude 70. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

D10-C2 · Choosing, ordering and complements · L3

Question: Choose a three-person committee from 4 women and 3 men. How many committees have at least one woman?

Answer: 34.

1. All committees: C(7,3)=35.
2. Subtract the one all-male committee.

Evidence anchors (0–3, descriptive only):

- 1 point: All committees: C(7,3)=35.
- 1 point: Subtract the one all-male committee.
- 1 point: conclude 34. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Counting an unordered committee as ordered; multiplying by a role count twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

D10-N3 · Last digits and power remainders · L3

Question: Find the last digit of 7^{2028}. Explain the cycle position.

Answer: 1.

1. The last digits cycle 7,9,3,1.
2. 2028 is a multiple of 4, so use the fourth position.

Evidence anchors (0–3, descriptive only):

- 1 point: The last digits cycle 7,9,3,1.

- 1 point: 2028 is a multiple of 4, so use the fourth position.
- 1 point: conclude 1. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

D10-L1 · Implication and counterexamples · L3

Question: For integers n, assess “6 divides n implies 3 divides n” and its converse. Give a proof or counterexample for each.

Answer: Original true; converse false, for example n=3.

1. n=6k is also 3(2k), proving the original.
2. The integer 3 satisfies divisibility by 3 but not by 6.

Evidence anchors (0–3, descriptive only):

- 1 point: n=6k is also 3(2k), proving the original.
- 1 point: The integer 3 satisfies divisibility by 3 but not by 6.
- 1 point: conclude Original true; converse false, for example n=3. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

C5-2 · Pigeonhole reasoning · L2

Question: Thirteen people are assigned birth months. What repeated-month fact is guaranteed?

Answer: At least two people share a birth month.

1. There are only 12 month categories.
2. Putting 13 people into 12 categories forces a repeated category.

Evidence anchors (0–3, descriptive only):

- 1 point: There are only 12 month categories.
- 1 point: Putting 13 people into 12 categories forces a repeated category.

- 1 point: give At least two people share a birth month. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Claiming a likely event is guaranteed; choosing boxes that do not force the desired pair
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Choose categories so that two objects in one category have the required property.

Follow-up: If the reasoning is uncertain, use C5-R without revealing this solution. If independent reasoning is sound, consider C5-4. Do not infer a gap from an unreached item.

C5-1 · Pigeonhole reasoning · L1

Question: If N objects are placed in k boxes, with N,k positive integers, what lower bound can you guarantee for the largest box occupancy? Explain.

Answer: At least ceil(N/k).

1. If every box held fewer than ceil(N/k), the total capacity would be less than N.
2. Thus at least one box meets the bound; no assumption of random placement is needed.

Evidence anchors (0–3, descriptive only):

- 1 point: If every box held fewer than ceil(N/k), the total capacity would be less than N.
- 1 point: Thus at least one box meets the bound; no assumption of random placement is needed.
- 1 point: give At least ceil(N/k). with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Claiming a likely event is guaranteed; choosing boxes that do not force the desired pair
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Choose categories so that two objects in one category have the required property.

Follow-up: If the reasoning is uncertain, use C5-R without revealing this solution. If independent reasoning is sound, consider C5-4. Do not infer a gap from an unreached item.

Reserve solutions

These are optional branches, not an extra homework set. Use at most two per session.

A1-4 · Equivalent equations and domain · L4

Question: Solve over the reals: $\sqrt{(x + 6)} = x$. Explain how you reject any extra candidate.

Answer: x = 3.

1. The square root is nonnegative, so $x \geq 0$.
2. Squaring gives $x^2 - x - 6 = 0$, so x is 3 or −2.
3. Only 3 is nonnegative and satisfies the original equation.

Evidence anchors (0–3, descriptive only):

- 1 point: The square root is nonnegative, so $x \geq 0$.
- 1 point: Squaring gives $x^2 - x - 6 = 0$, so x is 3 or −2.
- 1 point: give x = 3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

A1-R · Equivalent equations and domain · L2

Question: Solve $x(x + 2) = 0$. A student divides by x and reports only −2. What is missing?

Answer: x = 0 is also a solution.

1. Factor form already gives x = 0 or −2.
2. Dividing by x silently excludes zero.

Evidence anchors (0–3, descriptive only):

- 1 point: Factor form already gives x = 0 or −2.
- 1 point: Dividing by x silently excludes zero.
- 1 point: give x = 0 is also a solution. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling a possibly zero factor; accepting a value outside the original domain
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try the candidate in the original equation, including its denominator or square root.

Follow-up: If the reasoning is uncertain, use A1-R without revealing this solution. If independent reasoning is sound, consider A1-4. Do not infer a gap from an unreached item.

A2-4 · Polynomial roots and standard theorems · L4

Question: P is monic with integer coefficients and P(0)=−1, P(1)=1. Can P have a rational root? Justify.

Answer: No.

1. A rational root of a monic integer polynomial is an integer dividing its constant term, so it could only be ±1.
2. P(1) ≠ 0 rules out 1. Modulo 2, evaluating at −1 and 1 gives the same parity.
3. P(−1) is odd because P(1)=1 is odd, so P(−1) cannot be zero.

Evidence anchors (0–3, descriptive only):

- 1 point: A rational root of a monic integer polynomial is an integer dividing its constant term, so it could only be ±1.
- 1 point: P(1) ≠ 0 rules out 1. Modulo 2, evaluating at −1 and 1 gives the same parity.
- 1 point: give No. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating the rational-root list as a guarantee of roots; confusing P(a) with P(−a)
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: For division by x−a, think about substituting x=a. For rational roots, inspect the leading and constant coefficients.

Follow-up: If the reasoning is uncertain, use A2-R without revealing this solution. If independent reasoning is sound, consider A2-4. Do not infer a gap from an unreached item.

A2-R · Polynomial roots and standard theorems · L2

Question: For $3x^2 - 5x + 2$, test x=1 and use the result to factor the polynomial.

Answer: $(x - 1)(3x - 2)$.

1. P(1)=0, so x − 1 is a factor.
2. Multiplication confirms $(x - 1)(3x - 2) = 3x^2 - 5x + 2$.

Evidence anchors (0–3, descriptive only):

- 1 point: P(1)=0, so x − 1 is a factor.
- 1 point: Multiplication confirms $(x - 1)(3x - 2) = 3x^2 - 5x + 2$.
- 1 point: give $(x - 1)(3x - 2)$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating the rational-root list as a guarantee of roots; confusing P(a) with P(−a)
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: For division by x−a, think about substituting x=a. For rational roots, inspect the leading and constant coefficients.

Follow-up: If the reasoning is uncertain, use A2-R without revealing this solution. If independent reasoning is sound, consider A2-4. Do not infer a gap from an unreached item.

A3-4 · Vieta formulas and symmetric root expressions · L4

Question: If r,s solve $x^2 - 5x + 3=0$, find $1/r^3 + 1/s^3$.

Answer: 80/27.

1. Neither root is zero because rs=3.
2. $r^3+s^3=(r+s)^3-3rs(r+s)=125-45=80$.
3. Divide by $(rs)^3=27$.

Evidence anchors (0–3, descriptive only):

- 1 point: Neither root is zero because rs=3.
- 1 point: $r^3+s^3=(r+s)^3-3rs(r+s)=125-45=80$.
- 1 point: give 80/27. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand $(x-r)(x-s)$, or expand a square of a sum.

Follow-up: If the reasoning is uncertain, use A3-R without revealing this solution. If independent reasoning is sound, consider A3-4. Do not infer a gap from an unreached item.

A3-R · Vieta formulas and symmetric root expressions · L2

Question: Roots u,v of $x^2 - 6x + 4$ satisfy what value of u^2+v^2?

Answer: 28.

1. u+v=6 and uv=4.
2. Their squared sum is 36−8=28.

Evidence anchors (0–3, descriptive only):

- 1 point: u+v=6 and uv=4.
- 1 point: Their squared sum is 36−8=28.
- 1 point: give 28. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand $(x-r)(x-s)$, or expand a square of a sum.

Follow-up: If the reasoning is uncertain, use A3-R without revealing this solution. If independent reasoning is sound, consider A3-4. Do not infer a gap from an unreached item.

A4-4 · Substitution and symmetry · L4

Question: Find the number of real solutions of $x^4 - 3x^3 + 4x^2 - 3x + 1 = 0$.

Answer: Exactly one real solution: x=1.

1. Zero is not a root, so divide by x^2.
2. Set $t = x + 1/x$ to obtain $t^2 - 3t + 2 = 0$, hence t=1 or 2.
3. For real x, t=1 is impossible; t=2 gives x=1 only.

Evidence anchors (0–3, descriptive only):

- 1 point: Zero is not a root, so divide by x^2.
- 1 point: Set $t = x + 1/x$ to obtain $t^2 - 3t + 2 = 0$, hence t=1 or 2.
- 1 point: conclude Exactly one real solution: x=1. and justify the final case/domain/equality.

Possible error hypotheses, not automatic diagnoses:

- Treating x^2 as x; losing the nonzero restriction or real feasibility of a substituted value
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Look for a repeated expression. If you replace it, record which real values the replacement can take.

Follow-up: If the reasoning is uncertain, use A4-R without revealing this solution. If independent reasoning is sound, consider A4-4. Do not infer a gap from an unreached item.

A4-R · Substitution and symmetry · L2

Question: If $x + 1/x = 4$, $x \neq 0$, find $x^2 + 1/x^2$.

Answer: 14.

1. Square to obtain $x^2 + 2 + 1/x^2 = 16$.
2. Subtract 2.

Evidence anchors (0–3, descriptive only):

- 1 point: Square to obtain $x^2 + 2 + 1/x^2 = 16$.
- 1 point: Subtract 2.
- 1 point: give 14. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating x^2 as x; losing the nonzero restriction or real feasibility of a substituted value
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Look for a repeated expression. If you replace it, record which real values the replacement can take.

Follow-up: If the reasoning is uncertain, use A4-R without revealing this solution. If independent reasoning is sound, consider A4-4. Do not infer a gap from an unreached item.

A5-4 · Exponents and logarithms · L4

Question: Solve $\log_2(x-1)+\log_2(x+1)=3$ over the reals.

Answer: x=3.

1. The original arguments require x>1.
2. Combine logarithms: $\log_2(x^2-1)=3$, so $x^2=9$.
3. Only x=3 satisfies the original domain.

Evidence anchors (0–3, descriptive only):

- 1 point: The original arguments require x>1.
- 1 point: Combine logarithms: $\log_2(x^2-1)=3$, so $x^2=9$.
- 1 point: give x=3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Combining unlike bases incorrectly; ignoring positive logarithm arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Express the quantities with one base, and record logarithm domains before transforming.

Follow-up: If the reasoning is uncertain, use A5-R without revealing this solution. If independent reasoning is sound, consider A5-4. Do not infer a gap from an unreached item.

A5-R · Exponents and logarithms · L2

Question: Solve $3^{x-2}=27$.

Answer: x=5.

1. Write $27=3^3$.
2. Equate $x-2=3$.

Evidence anchors (0–3, descriptive only):

- 1 point: Write $27=3^3$.
- 1 point: Equate $x-2=3$.
- 1 point: give x=5. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Combining unlike bases incorrectly; ignoring positive logarithm arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Express the quantities with one base, and record logarithm domains before transforming.

Follow-up: If the reasoning is uncertain, use A5-R without revealing this solution. If independent reasoning is sound, consider A5-4. Do not infer a gap from an unreached item.

I1-4 · Signs and inequality transformations · L4

Question: Find all real x for which $x^2 \le 2x+3$ and $(x-2)/(x+1) > 0$ both hold.

Answer: $2 < x \le 3$.

1. The quadratic gives $-1 \le x \le 3$.
2. The rational inequality holds for $x < -1$ or $x > 2$, excluding -1.
3. Intersect the two sets.

Evidence anchors (0–3, descriptive only):

- 1 point: The quadratic gives $-1 \le x \le 3$.
- 1 point: The rational inequality holds for $x < -1$ or $x > 2$, excluding -1.
- 1 point: give $2 < x \le 3$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Failing to reverse order after negative division; multiplying by an expression of unknown sign
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Mark points where a factor is zero or undefined. Test one point in each interval.

Follow-up: If the reasoning is uncertain, use I1-R without revealing this solution. If independent reasoning is sound, consider I1-4. Do not infer a gap from an unreached item.

I1-R · Signs and inequality transformations · L2

Question: Solve $-2x \ge 8$. Explain the direction of the answer.

Answer: $x \le -4$.

1. Divide by negative 2.
2. Reverse ≥ to ≤.

Evidence anchors (0–3, descriptive only):

- 1 point: Divide by negative 2.
- 1 point: Reverse ≥ to ≤.
- 1 point: give $x \le -4$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Failing to reverse order after negative division; multiplying by an expression of unknown sign
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Mark points where a factor is zero or undefined. Test one point in each interval.

Follow-up: If the reasoning is uncertain, use I1-R without revealing this solution. If independent reasoning is sound, consider I1-4. Do not infer a gap from an unreached item.

I2-4 · Absolute value and intervals · L4

Question: For which real t does |x−1|+|x−5|=t have exactly two real solutions x?

Answer: t>4.

1. The minimum sum is 4, attained at every x in [1,5].
2. For t<4 there are no solutions.
3. For t>4 there is one solution to the left of 1 and one to the right of 5.

Evidence anchors (0–3, descriptive only):

- 1 point: The minimum sum is 4, attained at every x in [1,5].
- 1 point: For t<4 there are no solutions.
- 1 point: give t>4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Dropping one side of the absolute-value condition; counting an interval as one solution
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Locate the stated centres on a number line and interpret absolute values as distances.

Follow-up: If the reasoning is uncertain, use I2-R without revealing this solution. If independent reasoning is sound, consider I2-4. Do not infer a gap from an unreached item.

I2-R · Absolute value and intervals · L2

Question: Solve |x−4|≤2 and describe the interval in words.

Answer: 2≤x≤6.

1. The distance from 4 is at most 2.
2. The endpoints are 4−2 and 4+2.

Evidence anchors (0–3, descriptive only):

- 1 point: The distance from 4 is at most 2.
- 1 point: The endpoints are 4−2 and 4+2.
- 1 point: give 2≤x≤6. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Dropping one side of the absolute-value condition; counting an interval as one solution
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Locate the stated centres on a number line and interpret absolute values as distances.

Follow-up: If the reasoning is uncertain, use I2-R without revealing this solution. If independent reasoning is sound, consider I2-4. Do not infer a gap from an unreached item.

I3-4 · Bounds and equality cases · L4

Question: Positive x,y satisfy xy=4. Find the minimum of (x+y)(x+4y), with equality values.

Answer: 36 at $x=2\sqrt{2}$, $y=\sqrt{2}$.

1. Expand to $x^2+5xy+4y^2=x^2+4y^2+20$.
2. Since $xy=4$, $x^2+4y^2\geq 4xy=16$.
3. Equality requires $x=2y$; together with $xy=4$ this gives the stated values.

Evidence anchors (0–3, descriptive only):

- 1 point: Expand to $x^2+5xy+4y^2=x^2+4y^2+20$.
- 1 point: Since $xy=4$, $x^2+4y^2\geq 4xy=16$.
- 1 point: give 36 at $x=2\sqrt{2}$, $y=\sqrt{2}$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Applying AM–GM outside its domain; stating a bound without checking attainability
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Consider a nonnegative square. After a bound, check when equality can hold.

Follow-up: If the reasoning is uncertain, use I3-R without revealing this solution. If independent reasoning is sound, consider I3-4. Do not infer a gap from an unreached item.

I3-R · Bounds and equality cases · L2

Question: Positive a,b have a+b=8. Find the maximum ab.

Answer: 16 at a=b=4.

1. Nonnegative $(a-b)^2$ gives $4ab\leq 64$.
2. Equal values attain 16.

Evidence anchors (0–3, descriptive only):

- 1 point: Nonnegative $(a-b)^2$ gives $4ab\leq 64$.
- 1 point: Equal values attain 16.
- 1 point: give 16 at a=b=4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Applying AM–GM outside its domain; stating a bound without checking attainability
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Consider a nonnegative square. After a bound, check when equality can hold.

Follow-up: If the reasoning is uncertain, use I3-R without revealing this solution. If independent reasoning is sound, consider I3-4. Do not infer a gap from an unreached item.

F1-4 · Functions, domain and composition · L4

Question: For real x, f(x)=(x−1)/(x+1). Determine the domain of f(f(x)) and simplify it.

Answer: f(f(x))=−1/x for x≠−1,0.

1. The inner function requires x≠−1.
2. Its value must not equal −1; solving (x−1)/(x+1)=−1 gives x=0.
3. Simplifying the outer quotient yields −1/x, retaining both exclusions.

Evidence anchors (0–3, descriptive only):

- 1 point: The inner function requires x≠−1.
- 1 point: Its value must not equal −1; solving (x−1)/(x+1)=−1 gives x=0.
- 1 point: give f(f(x))=−1/x for x≠−1,0. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Reversing composition; forgetting the outer function can add a restriction
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Compute the inner output first and ask whether the outer function accepts it.

Follow-up: If the reasoning is uncertain, use F1-R without revealing this solution. If independent reasoning is sound, consider F1-4. Do not infer a gap from an unreached item.

F1-R · Functions, domain and composition · L2

Question: If f(x)=x+1 and g(x)=x², find f(g(3)) and g(f(3)).

Answer: 10 and 16.

1. g(3)=9, then f(9)=10.
2. f(3)=4, then g(4)=16.

Evidence anchors (0–3, descriptive only):

- 1 point: g(3)=9, then f(9)=10.
- 1 point: f(3)=4, then g(4)=16.
- 1 point: give 10 and 16. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Reversing composition; forgetting the outer function can add a restriction
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Compute the inner output first and ask whether the outer function accepts it.

Follow-up: If the reasoning is uncertain, use F1-R without revealing this solution. If independent reasoning is sound, consider F1-4. Do not infer a gap from an unreached item.

F2-4 · Reasoning from a functional equation · L4

Question: A function f:R→R satisfies f(x+y)=f(x)+f(y)+2xy for all real x,y, and f(1)=3. Determine f(r) for every rational r.

Answer: f(r)=r²+2r for every rational r.

1. Define g(x)=f(x)−x². Expanding (x+y)² shows g(x+y)=g(x)+g(y), and g(1)=2.
2. Additivity gives g(0)=0, g(−x)=−g(x), and g(p)=2p for every integer p. For a positive integer q, qg(p/q)=g(p)=2p.
3. Therefore g(p/q)=2p/q for every rational p/q, so f(r)=r²+2r.

Evidence anchors (0–3, descriptive only):

- 1 point: subtract x² and correctly establish additivity.
- 1 point: derive g(p/q)=2p/q, including zero and negative numerators.
- 1 point: conclude f(r)=r²+2r on the stated rational domain.

Possible error hypotheses, not automatic diagnoses:

- Assuming an unproved formula for f; failing to exploit substituted arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try zero, a negative input or replacing an input with its partner. Each substitution must be allowed.

Follow-up: If the reasoning is uncertain, use F2-R without revealing this solution. If independent reasoning is sound, consider F2-4. Do not infer a gap from an unreached item.

F2-R · Reasoning from a functional equation · L2

Question: If f(x+y)=f(x)+f(y) on R and f(1)=4, find f(3) and f(−1).

Answer: 12 and −4.

1. Repeated addition gives f(3)=3f(1).
2. Additivity at 1+(−1)=0 gives the negative value.

Evidence anchors (0–3, descriptive only):

- 1 point: Repeated addition gives f(3)=3f(1).
- 1 point: Additivity at 1+(−1)=0 gives the negative value.
- 1 point: give 12 and −4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Assuming an unproved formula for f; failing to exploit substituted arguments
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Try zero, a negative input or replacing an input with its partner. Each substitution must be allowed.

Follow-up: If the reasoning is uncertain, use F2-R without revealing this solution. If independent reasoning is sound, consider F2-4. Do not infer a gap from an unreached item.

L1-4 · Implication and counterexamples · L4

Question: Integers a,b satisfy: ab is divisible by 6. Must at least one of a,b be divisible by 6? Give a counterexample or proof, and explain which prime-divisibility fact still holds.

Answer: No: a=2,b=3. Each prime 2 and 3 divides at least one factor, but they can divide different factors.

1. The product 2·3 is divisible by 6 while neither factor is.
2. A prime dividing a product divides a factor, but 6 is composite and the prime factors need not occur in the same factor.

Evidence anchors (0–3, descriptive only):

- 1 point: The product 2·3 is divisible by 6 while neither factor is.
- 1 point: A prime dividing a product divides a factor, but 6 is composite and the prime factors need not occur in the same factor.
- 1 point: give No: a=2,b=3. Each prime 2 and 3 divides at least one factor, but they can divide different factors. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

L1-R · Implication and counterexamples · L2

Question: For integers n, is “3 divides n implies 6 divides n” true? Justify.

Answer: False; n=3 is a counterexample.

1. A counterexample must satisfy the premise.
2. Three is divisible by 3 but not by 6.

Evidence anchors (0–3, descriptive only):

- 1 point: A counterexample must satisfy the premise.
- 1 point: Three is divisible by 3 but not by 6.
- 1 point: give False; n=3 is a counterexample. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Proving the converse instead of the implication; offering an example that does not satisfy the premise
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Ask whether your example has a true premise and false conclusion. Label the direction proved.

Follow-up: If the reasoning is uncertain, use L1-R without revealing this solution. If independent reasoning is sound, consider L1-4. Do not infer a gap from an unreached item.

L2-4 · Quantifiers and negation · L4

Question: For real a, when is this true: for every real x there exists real y such that $y^2=x+a$? Negate the statement and decide it for every a.

Answer: It is false for every real a. Its negation is: there exists real x such that every real y has $y^2\neq x+a$. Choose $x=-a-1$.

1. A real square is nonnegative.
2. For any fixed a choose $x=-a-1$; the required square would be -1.
3. This gives a counterexample for every parameter a.

Evidence anchors (0–3, descriptive only):

- 1 point: A real square is nonnegative.
- 1 point: For any fixed a choose $x=-a-1$; the required square would be -1.
- 1 point: give It is false for every real a. Its negation is: there exists real x such that every real y has $y^2\neq x+a$. Choose $x=-a-1$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Swapping quantifier order; using one fixed witness when it must depend on an input
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the quantifiers in order and identify which choices may depend on earlier choices.

Follow-up: If the reasoning is uncertain, use L2-R without revealing this solution. If independent reasoning is sound, consider L2-4. Do not infer a gap from an unreached item.

L2-R · Quantifiers and negation · L2

Question: Negate “Every student solved at least one question.” Keep the student/question domains unchanged.

Answer: At least one student solved no questions.

1. Negating every introduces at least one student.
2. Negating at least one solved question means every question was not solved by that student.

Evidence anchors (0–3, descriptive only):

- 1 point: Negating every introduces at least one student.
- 1 point: Negating at least one solved question means every question was not solved by that student.
- 1 point: give At least one student solved no questions. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Swapping quantifier order; using one fixed witness when it must depend on an input

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the quantifiers in order and identify which choices may depend on earlier choices.

Follow-up: If the reasoning is uncertain, use L2-R without revealing this solution. If independent reasoning is sound, consider L2-4. Do not infer a gap from an unreached item.

S1-4 · Arithmetic sequences and sums · L4

Question: A list of consecutive positive integers starts at 1 and has sum 210. Delete one number and the mean of the remaining numbers becomes 10. Which number was deleted?

Answer: 20.

1. The sum 210 gives $n(n+1)/2=210$, so $n=20$.
2. Nineteen remaining numbers with mean 10 sum to 190.
3. The removed number is $210-190=20$.

Evidence anchors (0–3, descriptive only):

- 1 point: The sum 210 gives $n(n+1)/2=210$, so $n=20$.
- 1 point: Nineteen remaining numbers with mean 10 sum to 190.
- 1 point: give 20. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one term count; confusing last term with number of terms
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List the first few indices and compare the first and last terms.

Follow-up: If the reasoning is uncertain, use S1-R without revealing this solution. If independent reasoning is sound, consider S1-4. Do not infer a gap from an unreached item.

S1-R · Arithmetic sequences and sums · L2

Question: Find $5+8+11+\cdots+32$.

Answer: 185.

1. There are $(32-5)/3+1=10$ terms.
2. Their mean is $(5+32)/2$, so the sum is 185.

Evidence anchors (0–3, descriptive only):

- 1 point: There are $(32-5)/3+1=10$ terms.
- 1 point: Their mean is $(5+32)/2$, so the sum is 185.
- 1 point: give 185. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one term count; confusing last term with number of terms

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List the first few indices and compare the first and last terms.

Follow-up: If the reasoning is uncertain, use S1-R without revealing this solution. If independent reasoning is sound, consider S1-4. Do not infer a gap from an unreached item.

S2-4 · Geometric sequences and sums · L4

Question: Positive integers m,n satisfy $1+2+\cdots+2^{n-1}=3(1+4+\cdots+4^{m-1})$. Find all ordered pairs (m,n).

Answer: All pairs (m,2m), where m is any positive integer.

1. The left sum is 2^n-1.
2. The right sum is $3(4^m-1)/(4-1)=4^m-1$.
3. Thus $2^n=4^m=2^{2m}$, so n=2m; every positive m works.

Evidence anchors (0–3, descriptive only):

- 1 point: The left sum is 2^n-1.
- 1 point: The right sum is $3(4^m-1)/(4-1)=4^m-1$.
- 1 point: conclude All pairs (m,2m), where m is any positive integer. and justify the final case/domain/equality.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one exponent; dividing by r−1 when r=1; using a divergent infinite sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number of terms. Compare a finite sum with the same sum multiplied by its ratio.

Follow-up: If the reasoning is uncertain, use S2-R without revealing this solution. If independent reasoning is sound, consider S2-4. Do not infer a gap from an unreached item.

S2-R · Geometric sequences and sums · L2

Question: Find 1+2+4+8+16+32.

Answer: 63.

1. This is a six-term geometric sum with ratio 2.
2. Its sum is $2^6-1=63$.

Evidence anchors (0–3, descriptive only):

- 1 point: This is a six-term geometric sum with ratio 2.
- 1 point: Its sum is $2^6-1=63$.
- 1 point: give 63. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Off-by-one exponent; dividing by r−1 when r=1; using a divergent infinite sum

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number of terms. Compare a finite sum with the same sum multiplied by its ratio.

Follow-up: If the reasoning is uncertain, use S2-R without revealing this solution. If independent reasoning is sound, consider S2-4. Do not infer a gap from an unreached item.

S3-4 · Integer, square and cube sums · L4

Question: Without assuming the sum-of-squares formula, derive it by summing $(k+1)^3-k^3$ for $k=1,\ldots,n$. You may use $\Sigma k=n(n+1)/2$.

Answer: $\Sigma k^2=n(n+1)(2n+1)/6$.

1. The left side telescopes to $(n+1)^3-1$.
2. Expanding each difference gives $3\Sigma k^2+3\Sigma k+n$.
3. Substitute the integer sum and solve: $3\Sigma k^2=(n+1)^3-1-3n(n+1)/2-n$.

Evidence anchors (0–3, descriptive only):

- 1 point: The left side telescopes to $(n+1)^3-1$.
- 1 point: Expanding each difference gives $3\Sigma k^2+3\Sigma k+n$.
- 1 point: give $\Sigma k^2=n(n+1)(2n+1)/6$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

S3-R · Integer, square and cube sums · L2

Question: Find $1^2+2^2+\cdots+5^2$, and separately find $1+2+\cdots+5$.

Answer: 55 and 15.

1. The square terms sum to $1+4+9+16+25=55$.
2. The integer terms sum to $5\cdot6/2=15$; distinguish the two tasks.

Evidence anchors (0–3, descriptive only):

- 1 point: The square terms sum to $1+4+9+16+25=55$.
- 1 point: The integer terms sum to $5\cdot6/2=15$; distinguish the two tasks.
- 1 point: give 55 and 15. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using the integer-sum formula for squares; confusing sum of cubes with cube of a sum

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Expand the summand and separate the familiar sums. For a derivation, write the consecutive differences.

Follow-up: If the reasoning is uncertain, use S3-R without revealing this solution. If independent reasoning is sound, consider S3-4. Do not infer a gap from an unreached item.

S4-4 · Differences and telescoping · L4

Question: Let $a_n=n(n+1)(n+2)$. Find $a_{n+1}-a_n$ and hence evaluate $\Sigma\, k(k+1)$ for $k=1,\ldots,n$.

Answer: Difference $3(n+1)(n+2)$; sum $n(n+1)(n+2)/3$.

1. Factor the difference as $(n+1)(n+2)[(n+3)-n]=3(n+1)(n+2)$.
2. Thus $3k(k+1)=a_k-a_{k-1}$.
3. Sum from 1 to n, with $a_0=0$.

Evidence anchors (0–3, descriptive only):

- 1 point: Factor the difference as $(n+1)(n+2)[(n+3)-n]=3(n+1)(n+2)$.
- 1 point: Thus $3k(k+1)=a_k-a_{k-1}$.
- 1 point: give Difference $3(n+1)(n+2)$; sum $n(n+1)(n+2)/3$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling endpoints incorrectly; overlooking a shifted index
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the first three terms and last two terms, keeping their signs.

Follow-up: If the reasoning is uncertain, use S4-R without revealing this solution. If independent reasoning is sound, consider S4-4. Do not infer a gap from an unreached item.

S4-R · Differences and telescoping · L2

Question: Find $(1-1/2)+(1/2-1/3)+(1/3-1/4)$.

Answer: 3/4.

1. Cancel equal interior terms with opposite signs.
2. The endpoints give $1-1/4$.

Evidence anchors (0–3, descriptive only):

- 1 point: Cancel equal interior terms with opposite signs.
- 1 point: The endpoints give $1-1/4$.
- 1 point: give 3/4. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Cancelling endpoints incorrectly; overlooking a shifted index

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the first three terms and last two terms, keeping their signs.

Follow-up: If the reasoning is uncertain, use S4-R without revealing this solution. If independent reasoning is sound, consider S4-4. Do not infer a gap from an unreached item.

S5-4 · Arithmetic–geometric sums · L4

Question: For positive integer n, derive a formula for $\Sigma\, k\cdot 2^k$, k=1,…,n.

Answer: $(n-1)2^{n+1}+2$.

1. Let T be the sum and align 2T and T.
2. Their difference gives $T=n2^{n+1}-\Sigma 2^k$.
3. Since $\Sigma 2^k=2^{n+1}-2$, simplify to the stated formula.

Evidence anchors (0–3, descriptive only):

- 1 point: Let T be the sum and align 2T and T.
- 1 point: Their difference gives $T=n2^{n+1}-\Sigma 2^k$.
- 1 point: give $(n-1)2^{n+1}+2$. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating weighted terms as geometric; misaligning coefficients after shifting
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the sum and a ratio-multiple underneath it, aligning equal powers.

Follow-up: If the reasoning is uncertain, use S5-R without revealing this solution. If independent reasoning is sound, consider S5-4. Do not infer a gap from an unreached item.

S5-R · Arithmetic–geometric sums · L2

Question: Compute $1\cdot 3+2\cdot 3^2+3\cdot 3^3$.

Answer: 102.

1. The terms are 3,18,81.
2. Adding gives 102; identify both the linear weight and geometric power.

Evidence anchors (0–3, descriptive only):

- 1 point: The terms are 3,18,81.
- 1 point: Adding gives 102; identify both the linear weight and geometric power.
- 1 point: give 102. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Treating weighted terms as geometric; misaligning coefficients after shifting

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the sum and a ratio-multiple underneath it, aligning equal powers.

Follow-up: If the reasoning is uncertain, use S5-R without revealing this solution. If independent reasoning is sound, consider S5-4. Do not infer a gap from an unreached item.

C1-4 · Cases, product rule and completeness · L4

Question: How many six-digit positive integers use each of 0,1,2,3,4,5 exactly once and are divisible by 5?

Answer: 216.

1. The last digit must be 0 or 5.
2. If it is 0, permute the other five digits in 5!=120 ways. If it is 5, choose the first digit from 1,2,3,4, then arrange the remaining four digits in 4!=24 ways, giving 96.
3. The disjoint cases give 120+96=216.

Evidence anchors (0–3, descriptive only):

- 1 point: The last digit must be 0 or 5.
- 1 point: If it is 0, permute the other five digits in 5!=120 ways. If it is 5, choose the first digit from 1,2,3,4, then arrange the remaining four digits in 4!=24 ways, giving 96.
- 1 point: conclude 216. and justify the final case/domain/equality.

Possible error hypotheses, not automatic diagnoses:

- Allowing a leading zero; multiplying nonconstant continuation counts without splitting cases
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Specify exactly one completed object. Check whether every first choice has the same number of continuations.

Follow-up: If the reasoning is uncertain, use C1-R without revealing this solution. If independent reasoning is sound, consider C1-4. Do not infer a gap from an unreached item.

C1-R · Cases, product rule and completeness · L2

Question: A shirt has 4 colour choices and 3 size choices; every combination exists. How many combinations?

Answer: 12.

1. Each colour has three size choices.
2. Multiply 4 by 3.

Evidence anchors (0–3, descriptive only):

- 1 point: Each colour has three size choices.
- 1 point: Multiply 4 by 3.
- 1 point: give 12. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Allowing a leading zero; multiplying nonconstant continuation counts without splitting cases
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Specify exactly one completed object. Check whether every first choice has the same number of continuations.

Follow-up: If the reasoning is uncertain, use C1-R without revealing this solution. If independent reasoning is sound, consider C1-4. Do not infer a gap from an unreached item.

C2-4 · Choosing, ordering and complements · L4

Question: From 6 women and 5 men, choose a committee of 4 with at least 2 women, then appoint one committee member as chair. How many outcomes?

Answer: 1060.

1. The possible gender splits are 2+2, 3+1 and 4+0.
2. Count 15·10+20·5+15=265 committees.
3. Each has 4 possible chairs, giving 1060.

Evidence anchors (0–3, descriptive only):

- 1 point: The possible gender splits are 2+2, 3+1 and 4+0.
- 1 point: Count 15·10+20·5+15=265 committees.
- 1 point: conclude 1060. and justify the final case/domain/equality.

Possible error hypotheses, not automatic diagnoses:

- Counting an unordered committee as ordered; multiplying by a role count twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

C2-R · Choosing, ordering and complements · L2

Question: From 5 distinct people choose a pair, then distinguish one as leader. How many outcomes?

Answer: 20.

1. Choose an unordered pair in C(5,2)=10 ways.
2. Each pair has two possible leaders.

Evidence anchors (0–3, descriptive only):

- 1 point: Choose an unordered pair in C(5,2)=10 ways.
- 1 point: Each pair has two possible leaders.
- 1 point: give 20. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Counting an unordered committee as ordered; multiplying by a role count twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Describe whether positions or people are distinguished. Count the complement only within the correct sample space.

Follow-up: If the reasoning is uncertain, use C2-R without revealing this solution. If independent reasoning is sound, consider C2-4. Do not infer a gap from an unreached item.

C3-4 · Overlapping cases and inclusion–exclusion · L4

Question: How many integers from 1 through 200 are divisible by at least one of 2,3,5?

Answer: 146.

1. Single counts are 100,66,40.
2. Subtract pair intersections 33,20,13.
3. Add the triple intersection floor(200/30)=6: 206−66+6=146.

Evidence anchors (0–3, descriptive only):

- 1 point: Single counts are 100,66,40.
- 1 point: Subtract pair intersections 33,20,13.
- 1 point: give 146. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Double-counting intersections; subtracting a triple overlap too often
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Place one object belonging to every relevant set into your formula and count its net contribution.

Follow-up: If the reasoning is uncertain, use C3-R without revealing this solution. If independent reasoning is sound, consider C3-4. Do not infer a gap from an unreached item.

C3-R · Overlapping cases and inclusion–exclusion · L2

Question: Twelve students play chess, 10 play football and 4 play both. How many play at least one?

Answer: 18.

1. Add the two membership counts to get 22.
2. Subtract the four double-counted students.

Evidence anchors (0–3, descriptive only):

- 1 point: Add the two membership counts to get 22.
- 1 point: Subtract the four double-counted students.
- 1 point: give 18. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Double-counting intersections; subtracting a triple overlap too often

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Place one object belonging to every relevant set into your formula and count its net contribution.

Follow-up: If the reasoning is uncertain, use C3-R without revealing this solution. If independent reasoning is sound, consider C3-4. Do not infer a gap from an unreached item.

C4-4 · Bijections and counting two ways · L4

Question: Prove $\Sigma\, k \cdot C(n,k) = n \cdot 2^{n-1}$, $k=0,\ldots,n$, for positive integer n, by counting pairs (S,p), where S is a subset of an n-person set and p belongs to S.

Answer: Both sides count a subset with one distinguished member.

1. Group by subset size k: choose S in C(n,k) ways, then p in k ways.
2. Or first choose p in n ways, then decide independently whether each other person belongs to S, giving 2^{n-1}.
3. Both count exactly the same pairs.

Evidence anchors (0–3, descriptive only):

- 1 point: Group by subset size k: choose S in C(n,k) ways, then p in k ways.
- 1 point: Or first choose p in n ways, then decide independently whether each other person belongs to S, giving 2^{n-1}.
- 1 point: give Both sides count a subset with one distinguished member. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Missing the inverse correspondence; counting each unordered pair twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Define what one object on each side represents, and explain how to recover one from the other.

Follow-up: If the reasoning is uncertain, use C4-R without revealing this solution. If independent reasoning is sound, consider C4-4. Do not infer a gap from an unreached item.

C4-R · Bijections and counting two ways · L2

Question: How many shortest paths from (0,0) to (2,2) use only right and up moves?

Answer: 6.

1. A path is a four-letter string with two U and two R moves.
2. Choose the two U positions in C(4,2)=6 ways.

Evidence anchors (0–3, descriptive only):

- 1 point: A path is a four-letter string with two U and two R moves.
- 1 point: Choose the two U positions in C(4,2)=6 ways.
- 1 point: give 6. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Missing the inverse correspondence; counting each unordered pair twice
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Define what one object on each side represents, and explain how to recover one from the other.

Follow-up: If the reasoning is uncertain, use C4-R without revealing this solution. If independent reasoning is sound, consider C4-4. Do not infer a gap from an unreached item.

C5-4 · Pigeonhole reasoning · L4

Question: Choose any six distinct integers from 1,…,10. Prove that two chosen numbers sum to 11.

Answer: The five pairs {1,10},{2,9},{3,8},{4,7},{5,6} force a complete pair among six choices.

1. Partition the ten allowed numbers into five disjoint complementary pairs.
2. At most five numbers could be chosen while taking at most one from each pair.
3. Six choices therefore include both members of a pair.

Evidence anchors (0–3, descriptive only):

- 1 point: Partition the ten allowed numbers into five disjoint complementary pairs.
- 1 point: At most five numbers could be chosen while taking at most one from each pair.
- 1 point: give The five pairs {1,10},{2,9},{3,8},{4,7},{5,6} force a complete pair among six choices. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Claiming a likely event is guaranteed; choosing boxes that do not force the desired pair
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Choose categories so that two objects in one category have the required property.

Follow-up: If the reasoning is uncertain, use C5-R without revealing this solution. If independent reasoning is sound, consider C5-4. Do not infer a gap from an unreached item.

C5-R · Pigeonhole reasoning · L2

Question: Nine objects go into four boxes. What occupancy must at least one box have?

Answer: At least 3.

1. Two per box could hold only 8 objects.
2. Thus one box has at least ceil(9/4)=3.

Evidence anchors (0–3, descriptive only):

- 1 point: Two per box could hold only 8 objects.
- 1 point: Thus one box has at least ceil(9/4)=3.
- 1 point: give At least 3. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Claiming a likely event is guaranteed; choosing boxes that do not force the desired pair
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Choose categories so that two objects in one category have the required property.

Follow-up: If the reasoning is uncertain, use C5-R without revealing this solution. If independent reasoning is sound, consider C5-4. Do not infer a gap from an unreached item.

N1-4 · Divisibility from place value · L4

Question: A number written in base 8 has digits $d_m \ldots d_0$. Derive a digit test for divisibility by 7 and one for divisibility by 9.

Answer: For 7 use the ordinary digit sum; for 9 use an alternating digit sum.

1. The number is Σd_k8^k.
2. Modulo 7, 8 acts like 1; modulo 9, it acts like −1.
3. Hence the respective digit combinations have the same divisibility as the number.

Evidence anchors (0–3, descriptive only):

- 1 point: The number is Σd_k8^k.
- 1 point: Modulo 7, 8 acts like 1; modulo 9, it acts like −1.
- 1 point: give For 7 use the ordinary digit sum; for 9 use an alternating digit sum. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Remembering a digit trick without its base-dependent explanation
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number as a sum of digits times powers of its base.

Follow-up: If the reasoning is uncertain, use N1-R without revealing this solution. If independent reasoning is sound, consider N1-4. Do not infer a gap from an unreached item.

N1-R · Divisibility from place value · L2

Question: Is the decimal number 513 divisible by 9? Explain without long division.

Answer: Yes, because 5+1+3=9.

1. The digit sum is 9.
2. The decimal number and this sum have the same remainder modulo 9.

Evidence anchors (0–3, descriptive only):

- 1 point: The digit sum is 9.
- 1 point: The decimal number and this sum have the same remainder modulo 9.
- 1 point: give Yes, because 5+1+3=9. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Remembering a digit trick without its base-dependent explanation
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Write the number as a sum of digits times powers of its base.

Follow-up: If the reasoning is uncertain, use N1-R without revealing this solution. If independent reasoning is sound, consider N1-4. Do not infer a gap from an unreached item.

N2-4 · Gcd and integer constraints · L4

Question: Find all ordered pairs of positive integers a,b with gcd(a,b)=6 and a+b=54.

Answer: (6,48),(12,42),(24,30),(30,24),(42,12),(48,6).

1. Write a=6u,b=6v with u+v=9 and gcd(u,v)=1.
2. Since gcd(u,9−u)=gcd(u,9), u can be 1,2,4,5,7,8.
3. Multiply each pair by 6.

Evidence anchors (0–3, descriptive only):

- 1 point: Write a=6u,b=6v with u+v=9 and gcd(u,v)=1.
- 1 point: Since gcd(u,9−u)=gcd(u,9), u can be 1,2,4,5,7,8.
- 1 point: give (6,48),(12,42),(24,30),(30,24),(42,12),(48,6). with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Solving over reals instead of integers; forgetting coprimality after dividing by the gcd
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Find a divisor common to both coefficients. When scaling by a gcd, state the new pair must be coprime.

Follow-up: If the reasoning is uncertain, use N2-R without revealing this solution. If independent reasoning is sound, consider N2-4. Do not infer a gap from an unreached item.

N2-R · Gcd and integer constraints · L2

Question: Find gcd(48,18), showing the remainder steps.

Answer: 6.

1. 48=2·18+12 and 18=12+6.
2. Then 12=2·6, so the gcd is 6.

Evidence anchors (0–3, descriptive only):

- 1 point: 48=2·18+12 and 18=12+6.
- 1 point: Then 12=2·6, so the gcd is 6.
- 1 point: give 6. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Solving over reals instead of integers; forgetting coprimality after dividing by the gcd
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Find a divisor common to both coefficients. When scaling by a gcd, state the new pair must be coprime.

Follow-up: If the reasoning is uncertain, use N2-R without revealing this solution. If independent reasoning is sound, consider N2-4. Do not infer a gap from an unreached item.

N3-4 · Last digits and power remainders · L4

Question: Find the last digit of $7^{3^{2026}}$. Explain how you locate the exponent in the relevant cycle.

Answer: 7.

1. Powers of 7 have period 4 in their last digit.
2. The exponent 3^{2026} leaves remainder 1 modulo 4 because 2026 is even.
3. Therefore the last digit is 7.

Evidence anchors (0–3, descriptive only):

- 1 point: Powers of 7 have period 4 in their last digit.
- 1 point: The exponent 3^{2026} leaves remainder 1 modulo 4 because 2026 is even.
- 1 point: conclude 7. and justify the final case/domain/equality.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

N3-R · Last digits and power remainders · L2

Question: Find the last digit of 3^{12} and explain the zero-remainder case.

Answer: 1.

1. The cycle is 3,9,7,1.
2. Twelve is a multiple of 4, so use the fourth digit, 1.

Evidence anchors (0–3, descriptive only):

- 1 point: The cycle is 3,9,7,1.
- 1 point: Twelve is a multiple of 4, so use the fourth digit, 1.
- 1 point: give 1. with the stated domain/conditions and a valid argument.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

N3-X1 · Last digits and power remainders · L3

Question: Find the last digit of $7^{2026}+3^{2026}$.

Answer: 8.

1. Each power has last digit 9 from a four-term cycle.
2. Add to obtain last digit 8.

Evidence anchors (0–3, descriptive only):

- 1 point: Each power has last digit 9 from a four-term cycle.
- 1 point: Add to obtain last digit 8.
- 1 point: conclude 8. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

N3-X2 · Last digits and power remainders · L4

Question: Find the last digit of $7^{2026}\cdot 3^{2026}$.

Answer: 1.

1. Each factor has last digit 9.
2. Their product has the last digit of 81, namely 1.

Evidence anchors (0–3, descriptive only):

- 1 point: Each factor has last digit 9.
- 1 point: Their product has the last digit of 81, namely 1.
- 1 point: conclude 1. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Using cycle position zero; reducing an exponent by the divisor instead of the cycle length

- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: List a few remainders until the pattern restarts. An exponent divisible by the period uses the last position.

Follow-up: If the reasoning is uncertain, use N3-R without revealing this solution. If independent reasoning is sound, consider N3-4. Do not infer a gap from an unreached item.

EX-VJ · Vieta formulas and symmetric root expressions · L4

Question: Optional enrichment, with no baseline penalty: positive integers $a \le b$ satisfy $a^2+b^2+1=3ab$. For $a>1$, show that $b'=3a-b$ is a positive integer less than a and that (b',a) also solves the equation. Handle $a=1$ separately. Explain why repeated reduction reaches $(1,1)$.

Answer: For $a>1$, $0<b'<a$; $a=1$ gives $b=1$ or 2; descent reduces every solution to $(1,1)$.

1. Treat b as a root of $t^2-3at+(a^2+1)=0$. Its other root $b'=3a-b=(a^2+1)/b$ is a positive integer, and root substitution gives another solution.
2. If $a>1$, equality $a=b$ would give $a^2=1$, impossible; so $b \ge a+1$. Thus $b' \le (a^2+1)/(a+1)<a$.
3. For $a=1$, $b^2-3b+2=0$ gives $b=1$ or 2. The pair $(1,2)$ reduces to $(1,1)$. Otherwise replacing (a,b) by (b',a) strictly decreases the positive sum, so eventually $a=1$.

Evidence anchors (0–3, descriptive only):

- 1 point: Treat b as a root of $t^2-3at+(a^2+1)=0$. Its other root $b'=3a-b=(a^2+1)/b$ is a positive integer, and root substitution gives another solution.
- 1 point: If $a>1$, equality $a=b$ would give $a^2=1$, impossible; so $b \ge a+1$. Thus $b' \le (a^2+1)/(a+1)<a$.
- 1 point: conclude For $a>1$, $0<b'<a$; $a=1$ gives $b=1$ or 2; descent reduces every solution to $(1,1)$. with valid reasoning.

Possible error hypotheses, not automatic diagnoses:

- Wrong sign on the root sum; adding squares as the square of a sum
- A calculation or copying slip despite a valid setup; compare with a parallel item before concluding a concept gap.

Neutral hint, only after the independent attempt: Treat the equation as a quadratic in b, and find the other root using its sum and product.

Follow-up: Record prior exposure. If unfamiliar, record not encountered; do not assign a foundational gap or require a retry.